Highlights

Severe slugging with a four-equation two-fluid model and dynamic flow-pattern identification: well-posedness, discretisation, time integration and choke stability maps

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- Severe slugging simulated with the flow pattern identified along the whole system
- Low-void rule and virtual mass keep the two-fluid model hyperbolic where it is used
- Only conservative staggered finite volumes with donor cells carry the slug cycle
- Compiled right-hand side with exact Jacobians makes stability maps affordable
- Linearisation with the mass matrix predicts onset and period of slugging

Severe slugging with a four-equation two-fluid model and dynamic flow-pattern identification: well-posedness, discretisation, time integration and choke stability maps

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ARTICLE INFO

Keywords:
severe slugging
two-fluid model
well-posedness
flow patterns
finite volumes
automatic differentiation
choke

ABSTRACT

Severe slugging in pipeline–riser systems combines phase disappearance, moving liquid and gas fronts, changes of flow pattern and cycles of hundreds of seconds, which makes it a demanding test for transient two-fluid simulators. This work asks what a one-dimensional, four-equation two-fluid model with pattern-dependent closures needs to reproduce the cycle when the flow pattern is identified dynamically along the whole system, and uses the answer on a well–flowline–riser system of field dimensions (a 1500 m well, a 6 km flowline and a 335 m riser) with a topside choke. A characteristic analysis shows that the closures are ill-posed in large parts of their domain; a low-void rule, an objective virtual-mass term and an interfacial pressure in the intermittent pattern leave the model hyperbolic in all but 0.9 % of the states visited by a severe-slugging simulation, against 24 % without the virtual mass. The slug unit of the intermittent closure must take its film from the physics of the film rather than from the mass balance of the cell, or the unit is sold short at the triple point where the slug, the film and the cell hold the same gas. Only a staggered finite-volume scheme with phase-wise donor cells carries the cycle; integrating the non-conservative form loses liquid in one step per blowdown, whereas a conservative residual closes the mass balances to the Newton tolerance, and evaluating the face closures with the donor void fraction removes an odd–even mode and an unstable oscillation of the cells at the bend. A variable-order implicit integrator with physical bounds and range-scaled tolerances, driven by a compiled right-hand side with exact Jacobians, runs one hour of slugging in minutes. Linearised with the mass matrix of the conservative formulation, the steady state loses stability through a slow complex pair whose period is that of the simulated cycles: the onset of slugging is a Hopf bifurcation. On the field case: the cycle with the choke open, its period and the liquid surge; the opening that stabilises the riser and its production penalty; the effect of the gas–oil ratio of the well.

1. Introduction

Severe slugging is the cyclic blockage and blowout of a riser by liquid that accumulates at a low point of a production line (Schmidt, Brill and Beggs, 1980; Bøe, 1981; Pots, Bromilow and Konijn, 1987). When the gas in the line cannot carry the liquid up the riser, the liquid seals the riser base, the line pressurises while the riser fills, and once the pressure in the line overcomes the hydrostatic head the gas penetrates the riser and expels the liquid in a short burst, after which the liquid falls back and the cycle restarts (Taitel, 1986; Fabre, Peresson, Corteville, Odello and Bourgeois, 1990; Taitel, Vierkandt, Shoham and Brill, 1990). The cycle loads separators with liquid surges, causes pressure oscillations along the system and reduces production. Choking at the top of the riser can stabilise the flow (Schmidt, Brill and Beggs, 1979; Jansen, Shoham and Taitel, 1996), at the price of a back-pressure, which makes the stability boundary in the plane of choke opening and inflow conditions the central question for operation.

That boundary is a question of the stability of a steady state, and a few terms of the theory of dynamical systems will be used throughout (Strogatz, 2024; Kuznetsov, 2004). A steady state is *stable* if every small perturbation of it decays. Linearised about the steady state, a perturbation is a sum of modes that evolve as $e^{\lambda t}$, where the λ are the eigenvalues of the linearised system: the real part of an eigenvalue is the growth rate of its mode, the imaginary part its angular frequency, and the steady state is stable if all real parts are negative. As a parameter of the system changes

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— here the choke opening — the steady state can lose its stability where an eigenvalue crosses the imaginary axis; such a parameter value is a *bifurcation*. In a *Hopf bifurcation* the crossing is made by a pair of complex-conjugate eigenvalues, $\lambda = \sigma \pm \mathrm{i}\,\omega$ with σ changing sign and $\omega \neq 0$: beyond it a perturbation grows as an oscillation of period $2\pi/\omega$, until the non-linear terms bound it in a periodic oscillation of finite amplitude, a *limit cycle*, whose period near the bifurcation is $2\pi/\omega$. When the limit cycle grows from zero amplitude at the bifurcation (a supercritical bifurcation), the transition is gradual; when a cycle of finite amplitude already exists before it (subcritical), the stable steady state and the cycle coexist over a range of the parameter, and which one the system reaches depends on where it starts — a *hysteresis*. Further from the bifurcation the two eigenvalues of the pair may meet on the real axis and become real and positive. The oscillation is then a *relaxation oscillation* (van der Pol, 1926): a slow accumulation followed by a fast discharge, whose period is set by the accumulation rather than by an eigenvalue. A severe-slugging cycle, with its slow pressurisation of the line and its blowout, has this shape. Section 2 illustrates each of these terms on a model problem.

Models of severe slugging range from lumped descriptions of the cycle (Taitel et al., 1990; Sarica and Shoham, 1991) and low-order models for control (Storkaas and Skogestad, 2007; Jahanshahi and Skogestad, 2011), through drift-flux formulations and their nonlinear stability analysis (Baliño, Burr and Nemoto, 2010; Azevedo, Baliño and Andreolli, 2023), to one-dimensional two-fluid models of the kind used in commercial transient simulators (Bendiksen, Malnes, Moe and Nuland, 1991; Pauchon, Dhulesia, Binh-Cirlot and Fabre, 1994; Masella, Tran, Ferre and Pauchon, 1998), which have been compared with experiments in pipeline–riser systems and with choking (Malekzadeh, Henkes and Mudde, 2012; Park and Nydal, 2014; Li, Chen, Du and Han, 2022), to slug-capturing and slug-tracking studies (Issa and Kempf, 2003; Bonizzi, Andreussi and Banerjee, 2009; Kjeldby, Henkes and Nydal, 2013; Smith and Nydal, 2016). Han and Guo (2015) simulated terrain-induced severe slugging with a two-fluid model whose riser equations were made well-posed by an interfacial-pressure term, while the horizontal and descending sections were kept within the well-posed region of their closures. The two-fluid model is the natural framework, because it represents the slip between the phases, the fronts and the flow-pattern changes that a cycle involves, but it brings well-known difficulties: the model is not hyperbolic for every state and closure (Stuhmiller, 1977; Ramshaw and Trapp, 1978; López de Bertodano, Fullmer, Clausse and Ransom, 2017), phases appear and disappear, the closures change with the flow pattern, and the dynamics combine acoustic and void-wave time scales with a cycle of hundreds of seconds. Commercial codes resolve these difficulties in ways that are not published in full, and open studies usually prescribe the flow pattern of each section of the system.

This article reports what was needed to make a four-equation two-fluid model with pattern-dependent closures (Souza, 2010) simulate severe slugging with the flow pattern identified dynamically along the whole system, from the well inflow to the separator, and what can then be learnt from it about the stability of a well–pipeline–riser system with a topside choke. Section 2 first presents the method of analysis on a model problem with two linear equations and a single nonlinear boundary condition. The presentation then follows the chain of decisions in the order in which they matter: the model and its well-posedness (Sections 3 and 4), the spatial discretisation (Section 5), the time integration and what happens when its ingredients are removed (Section 6), the compiled backend (Section 7), and the application to a well–pipeline–riser system (Section 8). The contributions are: (i) a dynamic flow-pattern identification, smooth and evaluated on the local state, that can be used inside an implicit transient solver; (ii) a characteristic analysis by flow pattern, a regularisation built from physical terms, and a measure of well-posedness along the simulated trajectories rather than on a map; (iii) a comparison of collocated and staggered discretisations and of primitive and conservative time integration on the same severe-slugging problem, with exact mass balances, and the identification of an odd–even mode created by inconsistent face void fractions; (iv) an implicit integrator and a compiled right-hand side with exact Jacobians that make long transients of a well–flowline–riser system affordable; (v) the linear stability of the steady state of the discretised system, with the mass matrix that a conservative formulation requires, which predicts the onset and the period of the cycles; (vi) profiles, fronts and flow patterns along the cycle in a well–flowline–riser system of field dimensions, the stabilisation by the choke and the effect of the gas–oil ratio of the well. Section 9 collects the practical recommendations.

2. A model problem

The analysis that this article applies to severe slugging — a semi-discretisation in conservative variables, the linearisation of the steady state with its mass matrix, a continuation in the choke opening and transient simulations that test the linear predictions — can be followed step by step on a much simpler problem with the same structure. A line

$0 < x < 1$ carries a pressure p and a flux w governed by the telegraph equations,

$$C\frac{\partial p}{\partial t} + \frac{\partial w}{\partial x} = 0, \qquad L\frac{\partial w}{\partial t} + \frac{\partial p}{\partial x} = -R\,w, \tag{1}$$

two linear partial differential equations with constant coefficients, hyperbolic with wave speeds $\pm 1/\sqrt{LC}$. The line is fed at $x = 0$ by a linear source, $w(0,t) = k\,(P_r - p(0,t))$, and closed at $x = 1$ by an element with a single state y, the flux through it,

$$m\,\frac{\mathrm{d}y}{\mathrm{d}t} = p(1,t) - f(y) - a\,y, \qquad w(1,t) = y, \qquad f(y) = g_0 - \beta\,(y - y_0) + \tfrac{1}{3}(y - y_0)^3. \tag{2}$$

The line stands for the pipeline and its gas pocket, the source for the inflow from the reservoir, the end element for the riser and the resistance a for the choke. The characteristic f of the end element falls with the flux around y_0, as the head of a riser that lightens when more gas flows through it, and the cubic term bounds it; it is the only nonlinearity of the problem. A transmission line terminated by such a negative-resistance element is the tunnel-diode line of Nagumo and Shimura (1961), a close relative of the equations of FitzHugh (1961) and Nagumo, Arimoto and Yoshizawa (1962), and its oscillations are those of van der Pol (1926).

The line is discretised as the two-fluid model will be (Section 5): p at the centres of 50 cells, w at the faces, the inlet face given by the source and the outlet face by y. The semi-discrete right-hand side returns the time derivatives of the charges Cp, of the momenta Lw and of my, as the conservative formulation of the two-fluid model returns those of the phase masses, so that the linearisation about a steady state is the generalised eigenvalue problem $\mathbf{Jv} = \lambda\mathbf{Mv}$ with $\mathbf{M} = \mathrm{diag}(C, L, m)$. A lumped approximation shows what to expect. If the inertance and the friction of the line are neglected, the pressure is uniform along it, $p(x,t) \approx P(t)$, and integrating the first of Eqs. (1) over the line leaves two ordinary differential equations, for P and for the flux y through the end element:

$$C\,\frac{\mathrm{d}P}{\mathrm{d}t} = k\,(P_r - P) - y, \qquad m\,\frac{\mathrm{d}y}{\mathrm{d}t} = P - f(y) - a\,y. \tag{3}$$

At the steady state (P^*, y^*) the inflow equals the outflow, $k\,(P_r - P^*) = y^*$, and the pressure balances the end element, $P^* = f(y^*) + a\,y^*$. Small deviations δP and δy from it obey

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\delta P\\ \delta y\end{pmatrix} = \mathbf{B}\begin{pmatrix}\delta P\\ \delta y\end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix}-k/C & -1/C\\ 1/m & -r/m\end{pmatrix}, \qquad r = a + f'(y^*), \tag{4}$$

where r is the net differential resistance of the end element: the resistance a of the 'choke' plus the slope of the 'riser' characteristic, which is negative near y_0. The factors $1/C$ and $1/m$ are the inverse of the mass matrix of this two-state system, so that $\mathbf{B}$ is its $\mathbf{M}^{-1}\mathbf{J}$. The steady state is stable while the trace of $\mathbf{B}$, $-k/C - r/m$, is negative and its determinant, $(1 + k\,r)/(Cm)$, is positive. As a decreases, r decreases, and the trace vanishes at $r = -mk/C$: a pair of complex eigenvalues crosses the imaginary axis, a Hopf bifurcation in which the end element oscillates against the capacitance of the line with the frequency $\sqrt{\det\mathbf{B}} \approx 1/\sqrt{Cm}$. The two eigenvalues become real when the discriminant of $\mathbf{B}$, $(k/C - r/m)^2 - 4/(Cm)$, vanishes, at $r = mk/C - 2\sqrt{m/C}$; between the two values of r the steady state is an unstable focus, below the second an unstable node.

Fig. 1 follows the problem with $C = 10$, $L = R = 0.05$, $k = 0.5$, $m = 0.1$, $\beta = g_0 = y_0 = 1$ and P_r such that $y^* = 1$ at $a = 1$. A large resistance stabilises the line. As a decreases, the slow pair crosses the imaginary axis at $a_H = 0.976$ with a period of 6.75 (the lumped approximation gives 0.995 and 6.29), and its two eigenvalues meet on the real axis at $a_R = 0.756$ (0.792), below which two real global modes are unstable (Fig. 1b). The transients, integrated with the implicit formula of Section 6 from the steady state perturbed by 0.1 %, follow the spectrum. Above a_H the perturbation decays. Just below, it grows into a nearly sinusoidal cycle with the period of the pair (6.8 at $a = 0.95$, against 6.83 predicted), which widens as a decreases (Fig. 1a, d). Below a_R the cycle becomes a relaxation cycle: the line charges slowly while the flux through the end element stays on the left branch of f, the flux jumps to the right branch and the line discharges, with a flux wave that runs from the end into the line (Fig. 1e), and the flux jumps back, overshooting to negative values. Its period, 12.3 at $a = 0.5$, is set by the charging of the line, not by the pair, which no longer exists. In the order in which the choke opens, these are the stable flow, the pressure cycle and the severe slugging of Section 8, including the backflow through the choke at the end of a blowout.

The mass matrix matters. The eigenvalues of **J** alone, the Jacobian of the time derivatives of the charges and momenta, are those of a system in which C, L and m were all set to one: their largest real part stays at -0.025 for every a, and the line would be declared stable at every opening (Fig. 1b).

What the model problem does not have is the rest of this article: the characteristics of the two-fluid model can become complex, phases vanish, the masses are non-linear functions of the state (with a constant capacitance the primitive and conservative formulations coincide; with one that depends on the pressure, as for a gas, they do not), and the closures change with the flow pattern. The following sections take these up in the order of the introduction.

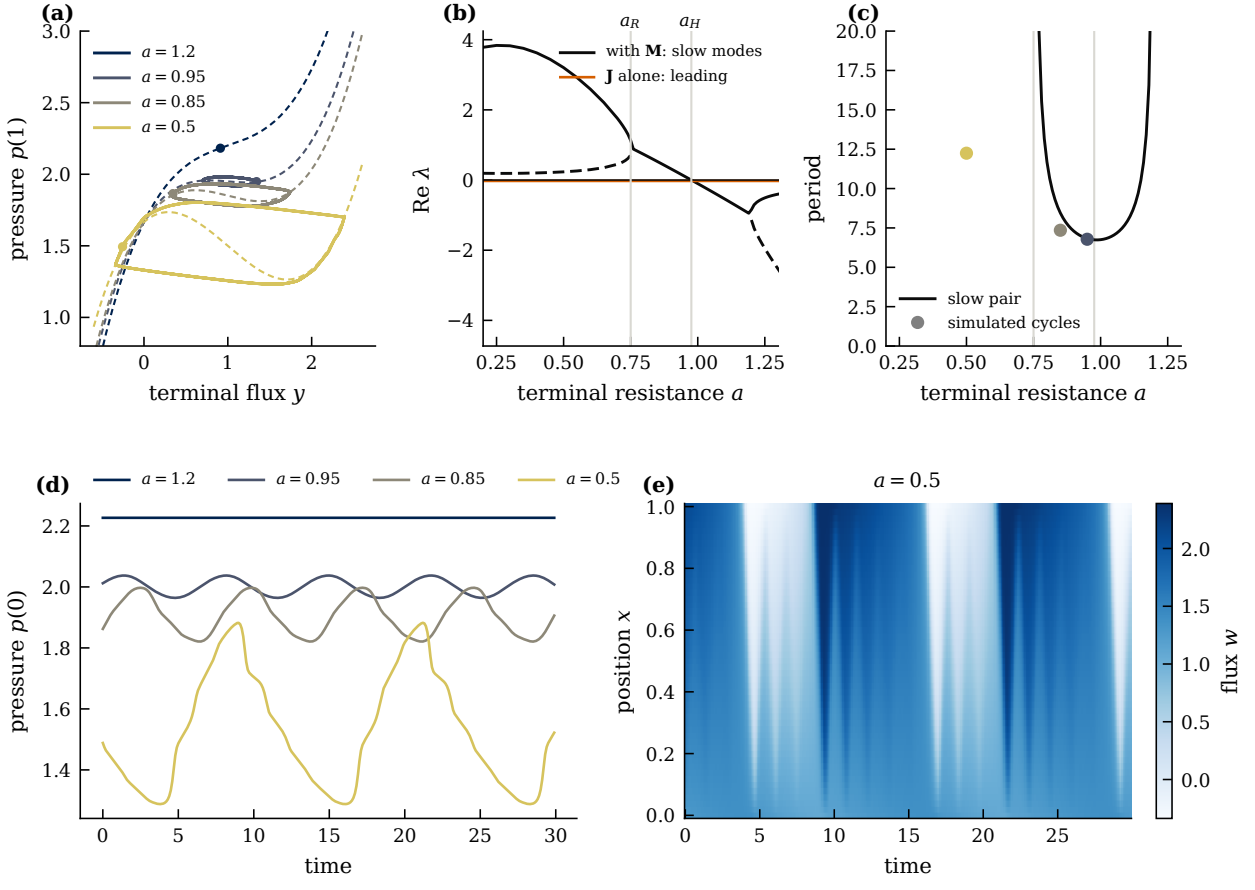


Figure 1: Model problem (1)–(2): (a) plane of the flux through the end element and the pressure at the end of the line, with the characteristic $f(y) + a\,y$ (dashed) and the simulated cycles (solid) for four values of a; (b) real part of the slow modes against a with the mass matrix, and of the leading eigenvalue of **J** alone; (c) period of the slow pair and of the simulated cycles; (d) pressure at the inlet of the line over the last 30 time units; (e) flux along the line in the relaxation cycle ($a = 0.5$).

3. Mathematical model

3.1. Two-fluid equations

The gas ($k = G$) and liquid ($k = L$) phases flow in a pipe of constant cross-section A and inclination $\theta(x)$. With the area fractions $\alpha_G + \alpha_L = 1$, the phase mass flow rates $q_k = \rho_k \alpha_k A v_k$ and a common pressure P, the mass and momentum balances are

$$\frac{\partial}{\partial t}\left(\rho_k \alpha_k A\right) + \frac{\partial q_k}{\partial x} = 0, \tag{5}$$

$$\frac{\partial q_k}{\partial t} + \frac{\partial}{\partial x}\left(q_k v_k\right) = -\alpha_k A \frac{\partial P}{\partial x} - A \frac{\partial}{\partial x}\left(\alpha_k \Delta P_k\right) - \Gamma_k - \rho_k \alpha_k A g \sin\theta + F_k^{vm}, \tag{6}$$

where ΔP_k is the difference between the average pressure of phase k and P (the hydrostatic level gradient in stratified flow, the interfacial pressure of dispersed flow), Γ_k the wall and interfacial shear forces per unit length and F_k^{vm} the virtual-mass force. Each phase follows a polytropic law, $\rho_k = \rho_{k0}(P/P_0)^{1/\gamma_k}$, with γ_G close to one for the gas and a large γ_L for the liquid. For the analysis the system is written in the primitive variables $\mathbf{y} = (\alpha_G, P, q_L, q_G)$ as

$$\frac{\partial \mathbf{y}}{\partial t} + \mathbf{A}(\mathbf{y})\,\frac{\partial \mathbf{y}}{\partial x} + \mathbf{b}(\mathbf{y}) = \mathbf{0}, \tag{7}$$

whose matrix $\mathbf{A}$ contains the convective terms, the pressure coupling through the phase compressibilities and the derivatives of ΔP_k with respect to α_G and P. The virtual-mass force acts in dispersed bubbly flow and is written in the objective form of Drew, Cheng and Lahey (1979),

$$F_G^{vm} = -F_L^{vm} = -C_{vm}\, w_B\, \alpha_G \rho_L A \left[\frac{D_G v_G}{Dt} - \frac{D_L v_L}{Dt} + (v_G - v_L)\left((\lambda - 2)\frac{\partial v_G}{\partial x} + (1-\lambda)\frac{\partial v_L}{\partial x} \right) \right], \tag{8}$$

with $C_{vm} = 0.5$, $\lambda = 1$ and w_B the weight of the bubbly pattern (Section 3.3).

3.2. Closure relations

The closures follow the flow pattern. In *stratified* flow the wetted perimeters and the hydrostatic pressure corrections follow from the liquid level (Taitel and Dukler, 1976), the wall friction from the Fanning factor of Churchill (1977) on each phase's hydraulic diameter, and the interfacial friction from Andritsos and Hanratty (1987). In *annular* flow the film friction follows Kowalski (1987) and the interfacial friction the same correlation on the film thickness. In *bubbly* flow the bubble diameter is limited by the Hinze–Calderbank breakup diameter (Hinze, 1955), the terminal velocity follows Mendelson (1967) and Fan and Tsuchiya (1990), the interfacial drag balances buoyancy at the terminal velocity, and the interfacial pressure is $\Delta P_L = 0.25\rho_L(v_G - v_L)^2$ (Stuhmiller, 1977). In *intermittent* flow a slug unit made of an aerated liquid slug and a stratified film region (Taitel and Barnea, 1990) weighs the bubbly and stratified closures by the lengths of the two regions, with the slug-body holdup of Barnea and Brauner (1985), the translational velocity of Bendiksen (1984) and the slug frequency of Zabaras (2000). The film region takes its gas fraction from the physics of the film, the stratified equilibrium of Taitel and Dukler (1976) in horizontal and downward sections and a Taylor bubble with a falling film (Taitel, Barnea and Dukler, 1980) in upward ones, and the slug fraction of the unit follows from the gas content of the cell, so that the unit always conserves the mass of the cell and reduces to the state of the cell in the liquid-rich and gas-rich limits; the slug frequency enters through the acceleration loss of the film picked up by the slug front (Taitel and Barnea, 1990). Appendix A gives the expressions. Where a phase vanishes, its velocity is undefined; Section 5 treats this numerically.

3.3. Dynamic flow-pattern identification

The flow pattern is identified at every cell and every time from stability criteria: the Kelvin–Helmholtz criterion for the stability of stratified flow in the direction of the pipe and the liquid-film trajectory criterion for reverse flow (Taitel and Dukler, 1976; Barnea, Shoham and Taitel, 1982), wind-generated waves for the wavy subpattern, the film-bridging and Froude criteria for annular flow (Taitel et al., 1980; Barnea, 1987), and the dispersion and coalescence criteria that separate bubbly from intermittent flow (Taitel et al., 1980; Barnea, 1987). The criteria, listed in Appendix B, are combined into weights w_S, w_A, w_B, w_I of the stratified, annular, bubbly and intermittent closures,

$$w_A = (1 - w_S)\,p_A, \qquad w_B = (1 - w_S)(1 - w_A)\,p_B, \qquad w_I = (1 - w_S)(1 - w_A)(1 - p_B), \tag{9}$$

so that $w_S + w_A + w_B + w_I = 1$ for any value of the indicators. The closure terms are the weighted sums of the four closures.

Steady-state flow-pattern maps compare superficial velocities with equilibrium solutions of the stratified and annular momentum balances; used as they are inside a transient solver they fail in three ways, which the present identification avoids. First, a sharp switch of the closures makes the residual non-smooth, and a Newton iteration then alternates between patterns from one iteration to the next; the indicators are therefore logistic functions whose width is a fraction of the critical value, $1/(1 + e^{-k(X - X_c)/|X_c|})$ with $k = 20$, so that criteria of very different magnitudes switch over the same relative interval. Second, the equilibrium level computed from superficial velocities does not describe a cell whose content is far from equilibrium: a gas pocket at rest in front of a liquid column has small superficial velocities and would be classified as bubbly flow. The criteria are therefore evaluated on the local state: the actual

Table 1
Ill-posed fraction of the void fraction–slip map by closure, without and with regularisation.

closure (inclination)	ill-posed (%)	max $\lvert\mathrm{Im}\,\lambda\rvert$ (m/s)	regularisation	ill-posed (%)
stratified (−2°)	47	5.31	low-void rule + virtual mass	7
intermittent (90°)	58	2.81	interfacial pressure, $\delta = 1.2$	0
bubbly (90°)	60	2.81	virtual mass, $C_{vm} = 0.5$	0
annular (90°)	93	3.00	interfacial pressure, $\delta = 1.2$	0
bubbly (90°)	60	2.81	virtual mass, $\lambda = 2$ (not objective)	99

Map of 70×71 states: $10^{-3} \le \alpha_G \le 0.995$, $-1 \le v_G - v_L \le 6$ m/s, $v_L = 0.5$ m/s, $P = 20$ bar, $D = 152.4$ mm.

level and void fraction replace the equilibrium ones in the stratified criteria, the bubbly criterion uses the larger of the drift-flux estimate and the cell void fraction, the film-bridging criterion of annular flow uses the larger of the equilibrium film fraction and the liquid fraction of the cell (a cell that holds more liquid than a bridging film cannot be annular, however small the superficial velocities), stratified flow is excluded in vertical sections and annular flow is assigned above $\alpha_G = 0.9$. Third, when the gas fraction is very small no pattern can be identified, since the criteria degenerate; below $\alpha_G = 0.01$ the closures blend smoothly into the bubbly ones (low-void rule; Section 4).

3.4. Well inflow, choke and separator

The well receives the reservoir fluids through a linear inflow performance relation, $q_k = J_k \max(0, P_{res} - P_{wf})$, with J_G/J_L the free-gas to liquid mass ratio of the produced fluid (Section 8.1), and the riser discharges through a choke into a separator at fixed pressure; the well and the flowline are two pipes of different diameters joined at the wellhead, where the flow rates of the well are imposed on the flowline inlet and the pressure of the flowline on the well outlet. The choke follows a homogeneous law: the mixture crosses the valve with the no-slip density weighted by volume, $\Delta P = \dot{m}\,Q/(Ka)^2$, with $\dot{m}$ and Q the mass and volumetric flow rates, $a \in (0, 1]$ the opening and Ka an effective flow area; the densities are evaluated at the downstream pressure. In the units of the simulator (ΔP in bar, $\dot{m}$ in $\mathrm{kg\,s^{-1}}$, Q in $\mathrm{m^3\,s^{-1}}$) $K = 0.0075\,c_v$ with the valve coefficient $c_v = 150$, which in SI units is an area of $35.6\,\mathrm{cm^2}$ for the fully open choke, against $182\,\mathrm{cm^2}$ for the pipe. A density weighted by mass flow, $\rho_M = (q_L\rho_L + q_G\rho_G)/(q_L + q_G)$, which is sometimes used, makes the choke almost transparent to gas, because the gas carries little mass, and becomes singular when the gas flows back through the choke at the end of a blowout and $q_L + q_G$ crosses zero; with it the choke does not stabilise the riser (Section 8.3).

4. Well-posedness

4.1. Characteristic analysis by flow pattern

The system (7) is hyperbolic when the eigenvalues of $\mathbf{A}$ are real; complex eigenvalues make the initial-value problem ill-posed, with growth rates proportional to the wavenumber, so that the numerical solution does not converge as the mesh is refined (López de Bertodano et al., 2017). Fig. 2 maps the largest imaginary part of the characteristic velocities over void fraction and slip for each closure at the conditions of the flowline of the application (20 bar, $D = 152\,\mathrm{mm}$, $v_L = 0.5\,\mathrm{m\,s^{-1}}$), and Table 1 gives the ill-posed fractions. Four regions are ill-posed. The stratified closure loses hyperbolicity when the gas layer becomes very thin, $\alpha_G \lesssim 0.005$, for any slip: the hydrostatic level gradient, which makes stratified flow hyperbolic up to the inviscid Kelvin–Helmholtz limit (Barnea and Taitel, 1994), vanishes when the level reaches the top of the pipe. This is precisely the state in which every liquid-filled cell is evaluated, since the void fraction of a cell without gas is cut at a small positive value. The bubbly closure is ill-posed at low void fraction with slip, the classical result of Stuhmiller (1977). The annular closure, with both phases at the same pressure, is ill-posed for almost any slip, and the intermittent closure at low void fraction and at large slip.

4.2. Regularisation

Three terms remove the ill-posed regions without changing the equations elsewhere. The low-void rule blends the closures of any pattern into the bubbly ones below $\alpha_G = 0.01$ with a C^1 ramp, which removes the thin-layer limit of the stratified closure; physically, a stratified gas layer of one per cent of the cross-section is not a flow pattern. The objective virtual-mass term (8) makes bubbly flow hyperbolic for every slip (Drew et al., 1979); with $\lambda = 2$, which keeps only the difference of material accelerations, the bubbly map becomes *more* ill-posed (99 % of the map against

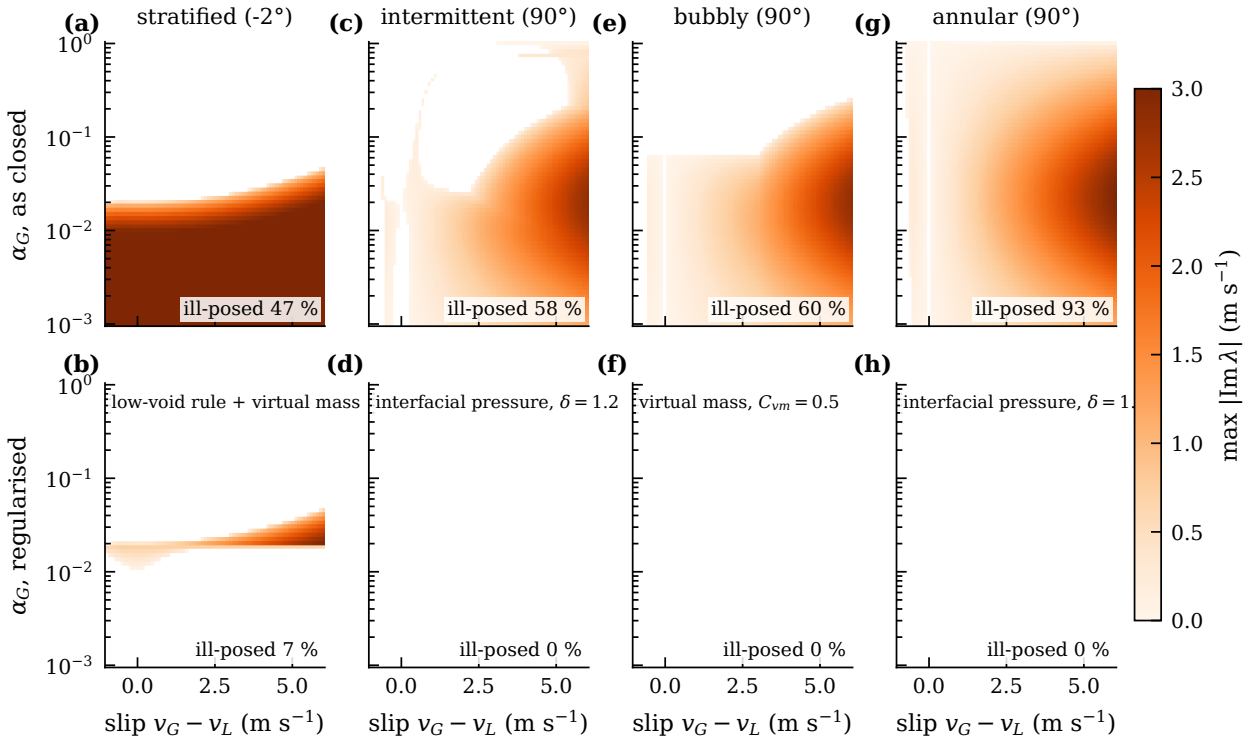


Figure 2: Largest imaginary part of the characteristic velocities over void fraction and slip, for the stratified, intermittent, bubbly and annular closures at 20 bar, $D = 152$ mm and $v_L = 0.5\,\mathrm{m\,s^{-1}}$: (a, c, e, g) as closed; (b, d, f, h) with the regularisation named in each panel. White: hyperbolic.

46 % without virtual mass), so the objective terms are essential. An interfacial-pressure term of the form proposed by Bestion (1990), $\Delta P_i = \delta\,\alpha_G\alpha_L\rho_G\rho_L(v_G - v_L)^2/(\alpha_G\rho_L + \alpha_L\rho_G)$ with $\delta \geq 1$, makes the annular and intermittent closures hyperbolic (Fig. 2d, h), but has no physical basis in those patterns and does not act on the thin stratified layer, where it is proportional to α_G. Surface tension adds a third-order dispersive term that limits the growth of capillary waves at millimetre scale; it does not change the characteristics and is irrelevant at cell sizes of metres.

4.3. Well-posedness along the simulated trajectories

A map shows where a closure is ill-posed, not whether a simulation goes there. The eigenvalues of **A** were therefore evaluated in every cell and at every output time of the simulations of Section 8. For the severe-slugging run with the choke open (8000 s, one state every 2 s), 0.9 % of the cell–time states are ill-posed with the low-void rule and the virtual mass. They are intermittent or annular cells (2.5 % and 2.0 % of the states with those patterns, against 0.4 % of the bubbly ones and none of the stratified ones), in the riser during the blowouts. Evaluated on the same states without the virtual mass, the fraction rises to 23.6 %, with 90 % of the bubbly cells and half of the riser states ill-posed, and without the low-void rule to 7.9 % (Table 2). The interfacial-pressure term of Bestion (1990) would leave 0.5 % (last row of Table 2); since it has no physical basis in the annular and intermittent patterns, it is not used in what follows. Without the virtual mass the simulation itself fails, after two blowouts (Section 6.3).

5. Spatial discretisation

5.1. Schemes

Five discretisations were applied to the same problem. Four are collocated, with the four primitive variables at the same nodes: the Galerkin and Petrov–Galerkin finite-element methods, and central and backward finite differences, the latter applied to all variables, the pressure included. The fifth is a staggered finite-volume scheme in the spirit of Harlow and Welch (1965) and Patankar (1980): void fraction and pressure at the cell centres, phase mass flow rates at the faces. The mass fluxes and the momentum flux of each phase use the donor cell defined by the sign of that phase's

Table 2
Ill-posed fraction of the cell–time states of a severe-slugging simulation, by dominant flow pattern, with the regularisation of the model and with the low-void rule or the virtual mass removed a posteriori on the same states.

closure set	all (%)	stratified	intermittent	bubbly	annular	riser
model (low-void rule + virtual mass)	0.9	0.0	2.5	0.4	2.0	0.9
without virtual mass	23.6	0.0	2.6	90.0	2.0	50.1
without low-void rule	7.9	0.0	2.6	28.1	2.0	0.9
without both	23.5	0.0	2.6	89.6	2.0	50.1
model + interfacial pressure	0.5	0.0	1.5	0.4	0.3	0.5

Ill-posed fraction of the cell–time states (one state every 5 s) of the severe-slugging run at an opening of 1 and 15 bar, overall, by dominant flow pattern of the model and in the riser; the variants are evaluated a posteriori on the same states.

flow rate, the pressure gradient is centred between neighbouring cells, and the pressure corrections ΔP_k are differenced between cell centres.

5.2. What a severe-slugging cycle demands

A cycle requires that a front between a liquid column and a gas pocket travel through the bend and up the riser, that a phase vanish from a cell and reappear, and that the inventory of both phases be conserved over many cycles. The collocated schemes of Section 5.1 have no dissipation on the void waves and develop node-to-node oscillations at the fronts; the backward difference applied to the pressure gradient of the primitive form is not conservative, and the Galerkin schemes fail at the first front, where the Newton iterations of the time step stop converging within a few seconds of simulated time. Only the staggered scheme completes the cycles, and only its conservative time integration (Section 5.5) keeps the inventories; Table 4 compares the two time formulations on the application case. On the offshore case the primitive integration of the same staggered scheme creates 1.7 % of the liquid and loses 1.9 % of the gas in one cycle (Table 4).

5.3. Vanishing phases

In a liquid-filled cell the gas momentum equation has no mass and the gas flow rate is undetermined. Three choices proved necessary. The mass flux of a phase through a face is multiplied by a C^1 limiter of the donor cell's fraction of that phase, so that a cell without a phase cannot give it away; this keeps the inventories non-negative without limiting the void fraction itself. A limiter on the void fraction, the obvious alternative, destroys mass whenever it acts and produces cycles that are artefacts of the lost liquid. The flow rate of the vanishing phase is relaxed, with a time constant of 0.05 s, to the value it would have without slip, computed with the donor cell's fraction floored at the small value at which the closures see the phase. Relaxing it to a zero flow rate instead leaves a virtual gas layer at rest above the moving liquid; a stratified closure evaluated in that cell sees a gas layer sheared by the liquid and exerts on a liquid-filled pipe an interfacial friction tens of times the wall friction, which slows a liquid column down. With the low-void rule of Section 4.2 the cell is evaluated with the bubbly closures and the two targets give the same cycle (Table 4); the no-slip target remains the one that does not depend on that rule. Finally, an artificial viscosity $\nu = Cch$ in the momentum equations, with c the phase sound speed, h the cell size and $C = 0.05$, damps the acoustic ringing of the short liquid columns that form at the riser base after a blowout, which otherwise forces time steps of milliseconds; it vanishes as $h \to 0$ and does not affect waves much longer than a cell.

5.4. Face void fraction and an odd–even mode

In the momentum equation of a face, the area fraction that multiplies the pressure gradient and the gravity term is that of the donor cell, while the closures (wall and interfacial friction) are naturally evaluated with the state at the face. If that state takes the arithmetic mean of the void fractions of the two neighbouring cells, the steady momentum balance of each face relates the donor fraction to the mean one; with the donor fraction α_i given, the balance fixes $(\alpha_i + \alpha_{i+1})/2$, so that $\alpha_{i+1} \approx 2\alpha^* - \alpha_i$ and the discrete steady state admits an odd–even mode. In the riser the void fraction then alternates from cell to cell with an amplitude that grows upwards. Evaluating the face closures with the donor void fraction, defined by the sign of the mixture volumetric flux, makes every term of the face balance use the same fraction: the steady profile becomes smooth, and an unstable oscillation localised in the cells of the bend disappears from the linearised steady state. The two choices give cycles that differ on the coarse mesh and converge with mesh refinement (Section 8.6).

5.5. Conservation

The staggered scheme conserves mass cell by cell in space, but conservation in time depends on the variables that the time integrator advances. Integrating the primitive form, $\partial\alpha_G/\partial t$ and $\partial P/\partial t$, with an implicit multistep formula does not conserve the phase masses, which are non-linear functions of α_G and P. The error is not uniform in time: it accumulates in steps, one per blowout, when a phase vanishes from many cells within a few seconds. On the application case the primitive formulation has, after 4000 s, created 1.7 % of the liquid fed and lost 1.9 % of the gas fed (Table 4), most of it in the blowout; the pressure range of the first cycle is the same, but the inventory that sets the following cycles is no longer the one the well delivered. The conservative formulation applies the multistep formula to the masses per unit volume, $\mathbf{T}(\mathbf{x}) = (\rho_G\alpha_G, \rho_L\alpha_L)$, while Newton's method still iterates on the primitive variables,

$$\mathbf{T}(\mathbf{x}_{n+1}) - \sum_j a_j\, \mathbf{T}(\mathbf{x}_{n-j}) - \gamma h\, \mathbf{f}(\mathbf{x}_{n+1}) = \mathbf{0}, \tag{10}$$

with the Jacobian $\partial\mathbf{T}/\partial\mathbf{x}$ evaluated analytically. The mass balances then close to the tolerance of the Newton iterations: on the application case the errors are 5×10^{-5} % of the liquid and 2×10^{-5} % of the gas fed after 4000 s, five orders of magnitude below the primitive formulation. The map $\mathbf{T}$ is singular at $\alpha_G = -\gamma_G/\gamma_L$, so the void fraction is allowed a small undershoot, $\alpha_G \geq -0.005$, well inside that limit. Verifying the balance itself requires care: integrating the boundary fluxes over the output times with the trapezoidal rule leaves an error of the order of a kilogram in each blowout, which is quadrature error and not loss of mass; the boundary fluxes must be integrated with the same multistep recurrence as the states.

6. Time integration

6.1. Integrator

The semi-discrete system is integrated by a backward differentiation formula of order one or two with variable step (Hairer and Wanner, 1996), a modified Newton method with the Jacobian reused while the convergence is fast, and an error estimate from the difference between the corrector and the predictor. Four features are not available in general-purpose stiff integrators and were needed here. The states are projected onto their physical bounds after each step, and the order falls back to one after a projection, since the history used by the second-order formula is inconsistent with a projected state. The absolute tolerance of each state is a fraction (10^{-2}) of the range that the state is expected to cover in the case (void fraction one, pressure up to the hydrostatic head of the riser, flow rates up to twenty times the liquid feed and fifty times the gas feed in a blowout), instead of a single value for states whose scales differ by four orders of magnitude; with a single tolerance the controller resolves the acoustic ringing of the liquid columns at a few per cent of the feed and the cost multiplies. The Newton iteration has a budget of twenty iterations, because its convergence is limited by the non-smoothness of the closures rather than by the quality of the Jacobian. The boundary fluxes are integrated with the same recurrence as the states, which gives an exact balance (Section 5.5).

Table 3 measures the Newton iteration on three windows of 400 s of the last cycle of the run with the choke open (Section 8.2): the blockage, the blowout and its end. The reference is the configuration of the application: a backtracking line search on the scaled residual while the correction is above the Newton tolerance (Dennis and Schnabel, 1996), and the local error estimated from the quadratic predictor with the Milne constant. Without the line search the steps grow by 40 to 75 %: in the blowout the controller then rejects a quarter of the attempted steps, all because the Newton iteration stalls or diverges from the predictor, which overshoots where a front crosses a cell; the exact, reused Jacobian is not the limitation. The line search must stop once the correction has converged: applied in every iteration, it refuses a converged correction whose residual does not fall — at the kinks of the closures the residual sits on a floor of the order of the correction itself — and the blowout window ends at the minimum step. The error estimate matters less in these windows: with the linear predictor and $r/(1 + 2r)\,(\mathbf{x} - \mathbf{x}^{pred})$, which is first order, the steps are within 7 % of the reference, because the step is limited by the Newton iteration and not by the error; on the model problem of Section 2, where the error limits the step, the Milne estimate saves 20 to 50 % of the steps at equal accuracy and restores the second-order scaling of the cost with the tolerance. Re-evaluating the Jacobian whenever the contraction is slow halves the steps of the blowout for 2.4 times the Jacobians, and starting the iteration from the previous state instead of the extrapolated predictor does not help.

6.2. Steady states, continuation and adaptive end time

Steady states are computed by Newton's method on $\mathbf{f}(\mathbf{x}) = \mathbf{0}$. From a generic initial estimate, the iteration does not converge for this problem; from a state reached by a transient that settles, or from the steady state of a neighbouring

Table 3
The Newton iteration of the integrator on three windows of the cycle with the choke open.

window	Newton	steps	rejected	Jacobians	right-hand sides
blockage, 13 794 s	reference (line search until convergence, Milne estimate)	82	25	33	1 381
blockage, 13 794 s	no line search	144	51	47	2 333
blockage, 13 794 s	line search in every iteration	134	43	58	2 178
blockage, 13 794 s	first-order error estimate	83	26	33	1 201
blockage, 13 794 s	Jacobian refreshed by rate	101	33	129	1 215
blockage, 13 794 s	previous state as initial iterate	90	30	37	1 510
blowout, 14 240 s	reference (line search until convergence, Milne estimate)	2 064	630	1 091	48 884
blowout, 14 240 s	no line search	3 511	1 172	1 451	78 803
blowout, 14 240 s	line search in every iteration	326 (failed)	115	136	5 588
blowout, 14 240 s	first-order error estimate	1 926	535	1 090	44 446
blowout, 14 240 s	Jacobian refreshed by rate	1 122	328	2 583	19 872
blowout, 14 240 s	previous state as initial iterate	2 079	645	1 196	49 862
end of blowout, 14 602 s	reference (line search until convergence, Milne estimate)	395	89	283	6 880
end of blowout, 14 602 s	no line search	551	168	296	10 186
end of blowout, 14 602 s	line search in every iteration	464	117	301	8 547
end of blowout, 14 602 s	first-order error estimate	385	85	286	7 295
end of blowout, 14 602 s	Jacobian refreshed by rate	755	209	1 804	12 379
end of blowout, 14 602 s	previous state as initial iterate	540	154	204	8 299

400 s of integration from the state of the run with the choke open at the given time (the last cycle of the run), with the settings of the application. Reference: modified Newton with the Jacobian reused while the convergence is fast, backtracking line search on the scaled residual while the correction is above the Newton tolerance, quadratic predictor with the Milne constant for the local error. No line search: step halved only when the residual grows after the first iteration. Line search in every iteration: also when the correction is already converged. First-order error estimate: linear predictor and $r/(1+2r)(\mathbf{x}-\mathbf{x}^{pred})$. Jacobian refreshed by rate: re-evaluated inside the iteration when the contraction per iteration is worse than 0.3, at most three times per step. Rejected: attempted steps rejected by the error control or by a failed Newton iteration. A run marked failed reached the minimum step. Wall time measured with other simulations running.

point, it converges in a few iterations, and continuation in the choke opening gives the steady state at every opening, stable or not. Pseudo-transient continuation is not used, because it converges only to stable equilibria (Kelley and Keyes, 1998). Transients that start from a steady state are perturbed by 0.1 % in pressure, and their time step is bounded by 1 s. Without that bound the error control lets the step grow to hundreds of seconds while the solution is still close to the equilibrium, and a backward differentiation formula with $|\lambda h| \gg 1$ damps an unstable mode as well as a stable one: the trajectory stays on an unstable equilibrium and the run is classified as stable. The transients of the application start instead from the blocked state of Section 8.1, which reaches the cycle directly. The simulated time is adaptive: the integration proceeds in windows of 2000 s and stops when the inlet pressure stays within 0.1 bar or when two consecutive windows have the same amplitude and period within 10 %, up to 8000 s. Even with the step bounded, a perturbation of 0.1 % of the pressure that grows at $10^{-3}\ \mathrm{s}^{-1}$ needs more than 2000 s, the length of a window, to exceed a range of 0.1 bar, so that a transient from the steady state, stopped by such a rule, misses the slowest instabilities near the onset; the stability of a steady state is better decided by its eigenvalues (Section 8.5).

6.3. What breaks without each ingredient

Table 4 removes one ingredient at a time from the configuration of the application, with the choke wide open at the base point, and reports whether 4000 s are completed, at what cost, with which mass error and with which cycle (the variants are defined in Table C.4). The reference completes the first cycle in 23 000 steps. The primitive time integration completes it with the same pressure range but with the mass errors of Section 5.5. Without the acoustic viscosity the integration crawls: five times as many steps for the same cycle. With the face closures evaluated at the mean void fraction of the two neighbouring cells the run fails at 3800 s, in the blowout, after five times as many steps as the reference. Remaining variants (runs in progress): low-void rule, virtual mass, bubble drag of the thesis, slug unit of the thesis (mass-consistent or not), slug frequency of the thesis, sharper switches, BDF1, uniform tolerance.

Table 4
Ablation: the application case (choke open at the base point, blocked start, 4000 s) without one ingredient at a time.

configuration	outcome	steps	wall time (s)	liquid error (%)	gas error (%)	blowdowns	interval (s)	P_{in} (bar)
reference (Section 8)	completed	23 332	1204	-4.8e-05	-2.3e-05	2	164	17.05–32.50
primitive time integration	completed	26 691	1249	+1.7e+00	-1.9e+00	1	–	17.00–32.62
no acoustic viscosity	completed	112 034	5496	-6.7e-04	+1.3e-04	0	–	17.55–33.15
face closures with the mean void fraction	fails at 3.77e+03 s	111 380	3494	–	–	0	–	18.93–34.29
no low-void rule	–	–	–	–	–	–	–	–
no virtual mass	completed	34 967	1783	-3.6e-04	+6.8e-05	0	–	17.05–33.81
bubble drag without g and σ/ρ_L	–	–	–	–	–	–	–	–
slug unit of the thesis, mass-consistent	–	–	–	–	–	–	–	–
slug unit of the thesis	–	–	–	–	–	–	–	–
slug frequency of the thesis	–	–	–	–	–	–	–	–
logistic switches with $k = 80$	–	–	–	–	–	–	–	–
first-order BDF	–	–	–	–	–	–	–	–
single absolute tolerance 10^{-6}	–	–	–	–	–	–	–	–

Mass errors: inventory change minus the boundary fluxes integrated with the integrator's own recurrence, at the end of the run, in per cent of the mass fed. Blowdowns: minima of the inlet pressure below the middle of its range (2nd to 98th percentile) after 300 s; interval: mean time between them. Wall-clock times with eight to twelve simulations sharing the machine; a run is stopped after 5400 s, 1.8 times the time of the reference.

Table 5
NumPy and JAX backends on the application case.

	NumPy	JAX
right-hand side (ms)	7.12	2.08
Jacobian (ms)	287.8 (coloured FD)	35.0 (exact)
compilation (s)	–	31
steps in 2000 s	–	–
wall time for 2000 s (s)	–	–

Application case with the choke open; 288 states, 29 colours in the finite-difference Jacobian; parity of the two right-hand sides 2e-13. Apple M5 Pro (measured with other simulations running).

7. Compiled backend

The fluid properties, closures, flow-pattern identification, schemes and network equations are written once against an injected array backend: the same code runs with NumPy and with JAX (Bradbury, Frostig, Hawkins, Johnson, Katariya, Leary, Maclaurin, Necula, Paszke, VanderPlas, Wanderman-Milne and Zhang, 2018), where the right-hand side is compiled just in time and its Jacobian is obtained exactly by forward-mode automatic differentiation (Griewank and Walther, 2008). The two backends agree to a relative 10^{-14}. Three rules make the physics differentiable and compilable: no in-place assignment, loops with a data-independent structure, and branches on values written as selections, with guards before functions with restricted domains so that tangents remain finite (a product $0 \times \infty$ in a tangent gives a NaN that only appears in the Jacobian). Table 5 compares the backends. The compiled right-hand side is three to four times faster than the NumPy one, and the exact Jacobian eight times faster than the coloured finite differences (Curtis, Powell and Reid, 1974) that need 29 evaluations of the right-hand side for the 288 states of the base mesh; 2000 s of severe slugging run in X instead of X (backend batch pending). The exact Jacobian does not reduce the number of Newton iterations, which is set by the non-smoothness of the closures, so the gain is in the cost per step. On a CPU the compiled code is limited by latency, not by throughput: the systems have one to a few hundred unknowns, and parallelism pays between simulations, which is how the batches of Section 8 were run.

8. Severe slugging in a well–pipeline–riser system

8.1. Case

The system is a production system in 300 m of water (Fig. 3, Table C.1): a vertical well of 1500 m (127 mm internal diameter) that receives the reservoir fluids through a linear inflow performance relation, a flowline of 2000 m on the sea bed (152.4 mm), the last 500 m of which descend at 2° to a low point 17.5 m below the wellhead, a circular bend of 5 m radius that turns the pipe from −2° to the vertical, a vertical riser of 317.5 m to the surface and a horizontal section of 10 m to the choke, which discharges into a separator at 10 bar. The well, the fluid, the diameters and the water depth are those of a case of the literature on compositional steady-state simulation of production systems (Carvalho, 2022),

Table 6
Cost of the transients of the application and of the steady states.

run	simulated (s)	steps	rejected	Jacobians	integration (s)	compilation (s)
opening 1	16000	20 800	6 655	11 108	449	incl.
opening 0.6	16000	11 291	3 977	5 179	270	incl.
opening 0.3	16000	9 271	3 343	4 370	229	incl.
opening 0.2	16000	21 019	6 727	6 787	369	incl.
opening 0.1	16000	12 777	4 489	4 496	255	incl.
GOR 30	16000	14 461	4 478	7 709	316	incl.
GOR 80	16000	42 021	12 698	23 715	1000	incl.
GOR 122.13	16000	42 754	13 731	21 449	959	incl.

Wall-clock time of the integration (JAX backend, Apple M5 Pro, five runs in parallel) and of the compilation of the right-hand side and the Jacobian, measured separately before the integration; *incl.*: runs made before the compilation was timed apart, where it is included in the integration time (a few seconds). Steady states by Newton with continuation in the opening (Section 8.5): 3.5 s per opening (median of 28, maximum 3.7 s), with the compiled right-hand side and the exact Jacobian.

with two changes: the final descent was added to the flowline, since a flowline that reaches the riser base horizontally does not hold a liquid seal, and the flowline was shortened from 6 km to 2 km. The period of severe slugging is set by the time to pressurise the gas in the flowline and is proportional to its volume; the shorter flowline gives cycles three times shorter at the same place in the severe-slugging region of the criterion of Bøe (1981), in which the superficial liquid velocity must exceed $P\,U_{SG}/(\rho_L g \alpha_G L)$: about $0.11\,\mathrm{m\,s^{-1}}$ for this flowline ($0.04\,\mathrm{m\,s^{-1}}$ for 6 km) against the 0.26–$0.31\,\mathrm{m\,s^{-1}}$ of the production during the blocked phase. The base mesh has 15 cells in the well and 35 in the flowline–riser pipe: 8 in the horizontal section, 5 in the descent, 4 in the bend, 16 in the riser and 2 at the top.

The fluid is the light oil of that case ($673\,\mathrm{kg\,m^{-3}}$ at standard conditions, 78° API) with its associated gas, described in the simulator as a polytropic pair fitted to a Peng–Robinson flash at 40 °C: the densities and viscosities at 20 bar, the pressure dependence of the gas fitted between 10 and 60 bar (Table C.2). Since the simulator has neither mass transfer nor an energy equation, the division of mass between the phases is frozen at the conditions of the flowline: the oil carries the solution gas of 20 bar and 40 °C, $R_s = 23.2\,\mathrm{m^3\,m^{-3}}$ at standard conditions, and the rest of the gas is free. The well is described with the variables of the field: a productivity index J of stock-tank oil, a reservoir pressure P_{res} and a gas–oil ratio GOR at standard conditions. With the standard densities of the oil, of the dissolved gas and of the free gas (ρ_o, ρ_{gd}, ρ_{gf}) the mass per standard cubic metre of oil is $m_o = \rho_o + \rho_{gd} R_s + \rho_{gf}(\mathrm{GOR} - R_s)$, the mass fraction of free gas is $w_G = \rho_{gf}(\mathrm{GOR} - R_s)/m_o$, and the inflow relation of Section 3.4 has $J_L = (1 - w_G)\,J m_o/86400$ and $J_G = w_G\,J m_o/86400$ in $\mathrm{kg\,s^{-1}\,bar^{-1}}$. More or less gas from the well is a higher or lower GOR with the same oil and the same free gas. The base point of the analysis is a weak well, $J = 5\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$, with $P_{res} = 200$ bar and $\mathrm{GOR} = 50\,\mathrm{m^3\,m^{-3}}$ ($w_G = 0.029$), which produces about $4\,\mathrm{kg\,s^{-1}}$: the flowline then runs stratified and the riser can hold a liquid seal against the reservoir, the two conditions of severe slugging. A productive well at the same pressure (the $42.9\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$ that reproduce the nominal point of the reference case) delivers 10 to $20\,\mathrm{kg\,s^{-1}}$, the flowline runs intermittent and the riser never blocks; a well at a lower reservoir pressure dies when the riser blocks, since the liquid columns of the riser and of the well exceed the reservoir pressure. Drier and wetter wells, up to the GOR of the reservoir fluid itself ($122\,\mathrm{m^3\,m^{-3}}$, $w_G = 0.099$), are explored in Section 8.4. The choke has $c_v = 150$, which gives a pressure drop below 1 bar at $5\,\mathrm{kg\,s^{-1}}$ when fully open.

The transients start from a blocked state constructed by the case generator (Appendix C): the flowline stratified at the equilibrium level of the nominal flow rates, the bend and the riser full of liquid, the well at the no-slip void fraction, the pressure marched from the separator with the local mixture density, and the total flow rate solved from the inflow relation. Filling the flowline from a no-slip start takes hours of simulated time; the blocked start reaches the first blowout within the first 300 s. All transients of this section start at $t = 0$ from this state and run for 16 000 s. The first 8000 s are a transient: with the choke open the system goes through five cycles of decreasing period, from 2050 to 1300 s, before settling from about 8600 s on the limit cycle described next; the signatures of Tables 8 and 9 are measured over the last 8000 s, Fig. 6 shows whole runs and Fig. 4 the last complete cycle of one of them. The only runs that start from a saved state are the windows of Table 3, which measure the integrator on three phases of that cycle.

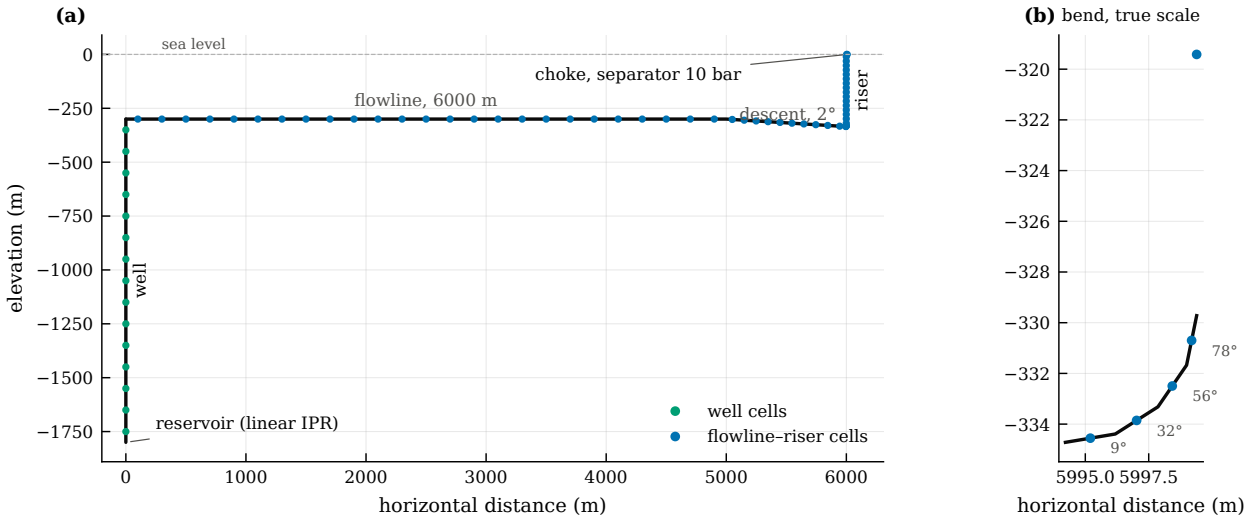


Figure 3: (a) Pipeline–riser system with the cell centres of the base mesh; (b) the bend at true scale with the inclination of each cell.

Table 7
Configuration of the application case.

component	setting
geometry	well 1500 m vertical ($D = 127$ mm); flowline 5000 m horizontal and 1000 m at $-2°$, bend of radius 5 m, riser 334.9 m, top 10 m ($D = 152.4$ mm)
mesh	15 cells in the well, 57 in the flowline–riser ($25 + 10 + 4 + 16 + 2$)
liquid	$\rho_{L0} = 646.7\,\mathrm{kg\,m^{-3}}$ at 20 bar, $\mu_L = 0.314$ mPa s, $\gamma_L = 1000$
gas	$\rho_{G0} = 15\,\mathrm{kg\,m^{-3}}$ at 20 bar, $\mu_G = 12.1$ μPa s, $\gamma_G = 1$, $\sigma = 0.0163\,\mathrm{N\,m^{-1}}$
well	$J = 5\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$, $P_{res} = 200$ bar, GOR $= 50\,\mathrm{m^3\,m^{-3}}$ ($w_G = 0.029$; $J_L = 0.041\,\mathrm{kg\,s^{-1}\,bar^{-1}}$, $J_G = 0.0012\,\mathrm{kg\,s^{-1}\,bar^{-1}}$)
choke	homogeneous law, $c_v = 150$, opening 0.1 to 1; separator at 10 bar
closures	dynamic flow pattern (smooth, local, $k = 20$), low-void rule below 1 %, $C_{vm} = 0.5$ ($\lambda = 1$), physical bubble drag, slug unit with the film at equilibrium
scheme	staggered finite volumes, conservative, donor face closures, no-slip relaxation with $\tau = 0.05$ s, $\nu = 0.05\,c\,h$
integrator	BDF of order 1–2, rtol 0.003, range-scaled atol 0.01, Newton budget 20, JAX
start	blocked: flowline stratified at equilibrium, bend and riser full of liquid

8.2. The cycle with the choke open

With the choke fully open the system does not settle. Fig. 4 shows the last complete cycle of the run, which has period two: blowouts of two sizes alternate, and the double cycle lasts about 3000 s (1250 and 1870 s in the cycle shown). At the end of the large blowout the riser is empty (its mean liquid holdup falls to 0.22 and the top of the liquid column is at the low point), the flowline inlet pressure is at its minimum of 14.7 bar and the liquid seal upstream of the low point is about 100 m long. The riser refills from its base over some 1100 s, in steps that follow the arrival of liquid slugs from the flowline, while the seal grows to 360 m and the inlet pressure rises almost linearly, by 12 bar per 1000 s: the gas produced by the well ($0.12\,\mathrm{kg\,s^{-1}}$) accumulates in a flowline that runs stratified over most of its length and intermittent near the seal, and the riser is a bubbly column that reaches the top some 300 s before the blowout. When the inlet pressure reaches 31 bar — the separator pressure plus the 20.1 bar of a liquid-filled column of 317 m plus the friction of the seal — the seal starts to move. The first blowout is partial: the liquid is pushed over the top at up to $33\,\mathrm{kg\,s^{-1}}$, ten times the mean production; the gas reaches the riser base and expels the upper two thirds of the column, whose top falls to about 100 m above the low point and whose mean holdup falls to 0.28; and the inlet pressure falls only to 22 bar, since the lower third of the column stays in place. The riser refills in some 450 s, the pressure rises again to 30 bar over 900 s, and the second blowout is complete: $45\,\mathrm{kg\,s^{-1}}$ of liquid and $1.1\,\mathrm{kg\,s^{-1}}$ of gas through the choke, the riser left annular at the top and then empty, the pressure back to 14.7 bar — which resets the cycle. The two blowouts take about a tenth of the double period. The mean production over the last 8000 s is $3.86\,\mathrm{kg\,s^{-1}}$, against the

3.60 kg s^{-1} that the well delivers in the steady state which the choke stabilises at $a = 0.1$ (Section 8.3): the cycle does not cost production on average, it concentrates it in surges of ten times the mean. Fig. 5 follows the cycle with profiles along the whole system at six instants, and Table 8 gives its signature.

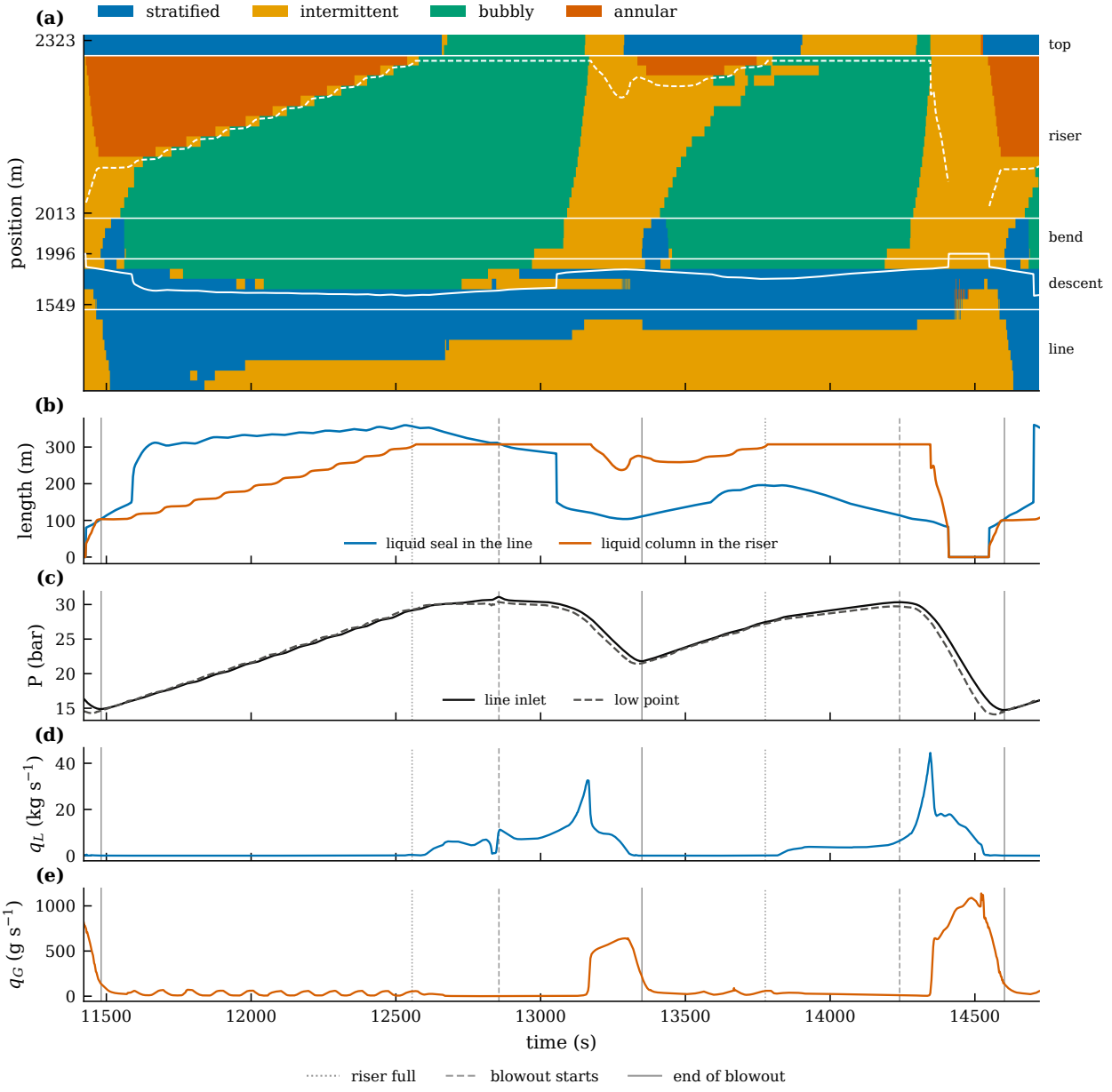


Figure 4: The last complete cycle of the run with the choke open at the base point (the double cycle from 11 480 s to 14 600 s of the 16 000 s run started from the blocked state): (a) dominant flow pattern along the flowline–riser pipe (one row per cell) with the end of the liquid seal in the flowline (solid) and the top of the liquid column in the riser (dashed); (b) the two fronts in metres; (c) pressure at the flowline inlet and at the low point; (d) liquid and (e) gas flow rates through the choke. Vertical lines: riser full, blowout starts, end of blowout.

8.3. Closing the choke

Fig. 6 shows the liquid holdup along the pipe against time for three openings, from the blocked start, with the flowline inlet pressure and the liquid outflow below, and Table 8 the signature of the cycles for the five openings of

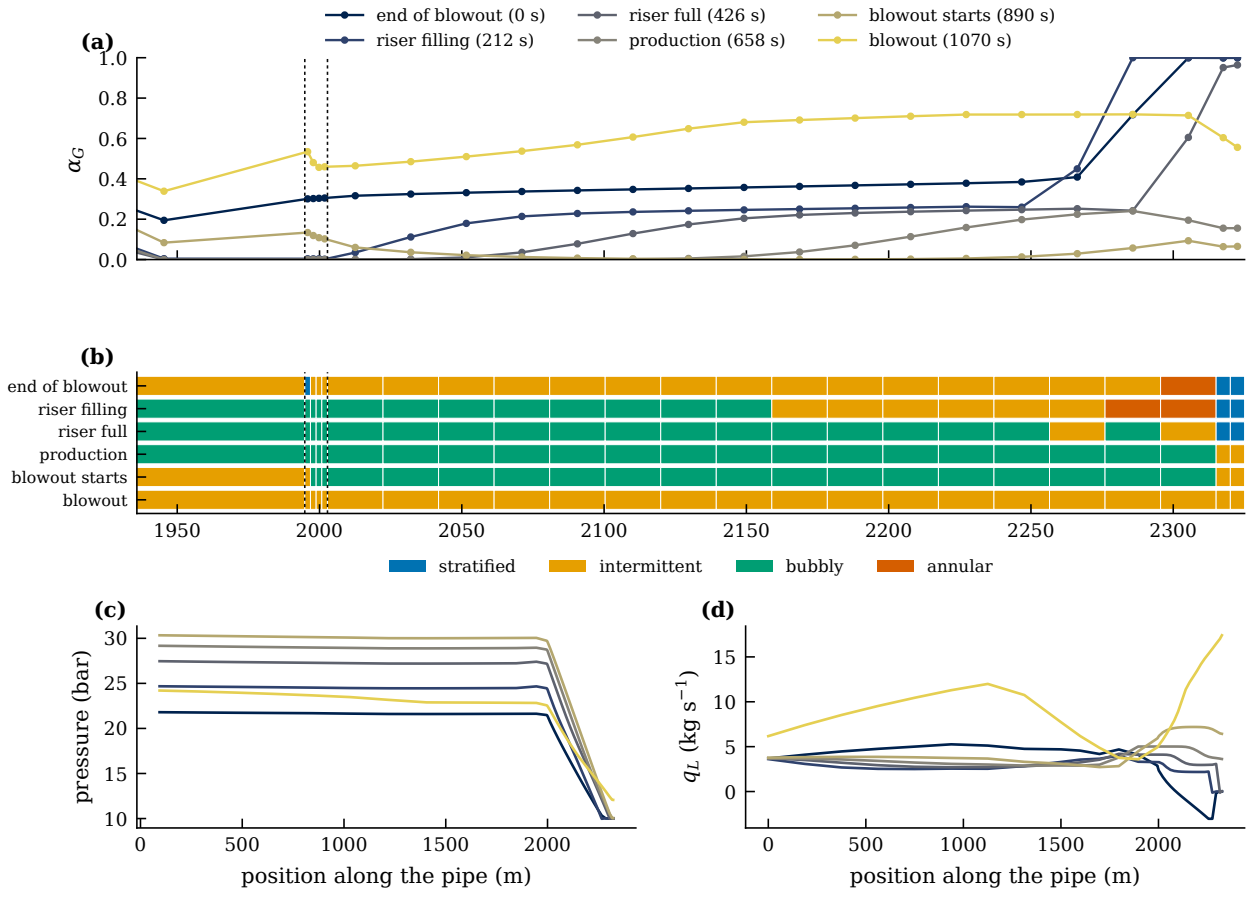


Figure 5: Profiles along the flowline–riser pipe at six instants of the last cycle with the choke open: void fraction and dominant flow pattern near the bend, pressure and liquid flow rate along the whole pipe.

the batch over the last 8000 s. Closing the choke to 0.6 keeps severe slugging but removes the period doubling: the cycle becomes regular, with a period of 1240 s, the pressure between 21.7 and 30.6 bar, a seal of 215 m and a surge of 32 kg s^{-1} (8.6 times the mean), and the riser no longer empties (its mean holdup stays above 0.54), because the choke, which takes up to 4 bar during the surge, holds back the column. At 0.3 the cycle changes character: the riser stays full of liquid (holdup above 0.64, the column always at the top), the seal is 120 m long, the period falls to 730 s and the inlet pressure oscillates between 25.7 and 28.5 bar; what leaves the riser is a pulse of 10 kg s^{-1}, 2.7 times the mean, in every cycle — a pressure cycle with liquid pulses rather than a blowout. At 0.2 the pressure cycle survives with the same period (720 s) and an amplitude of 0.7 bar, with the outflow smoothed to 1.2 times the mean; at 0.1 the system is steady at 33.0 bar, with the choke taking 5.8 bar. The opening that stabilises the riser is therefore between 0.1 and 0.2, where the choke takes 2 to 6 bar against the 20 bar of the riser column (Section 8.5 locates it), and the mean production changes little along the way: between 3.7 and 3.9 kg s^{-1} for openings from 1 to 0.2, and 3.60 kg s^{-1} at 0.1, where the choke adds its own back-pressure to the well.

8.4. More and less gas from the well

The gas–oil ratio of the well sets the gas that accumulates in the flowline during the blocked phase. Fig. 7 and Table 9 compare the base point (GOR 50, $w_G = 0.029$) with a drier well (GOR 30, barely above the solution gas, $w_G = 0.007$) and with wetter ones (GOR 80, $w_G = 0.059$, and the 122 m^3 m^{-3} of the reservoir fluid itself, $w_G = 0.099$), all with the choke open. The three wetter wells give the same severe slugging: the riser empties at every large blowout (mean holdup down to 0.17), the pressure swings between 15 and 31 bar and the surge reaches 40 to 45 kg s^{-1}, since both are set by the riser column and not by the gas; the gas rate sets the mean period, 1530 s with GOR 50 and 80 (both with period doubling) and 1180 s with GOR 122, and the mean production, which grows from 3.9 to 4.3 kg s^{-1} through

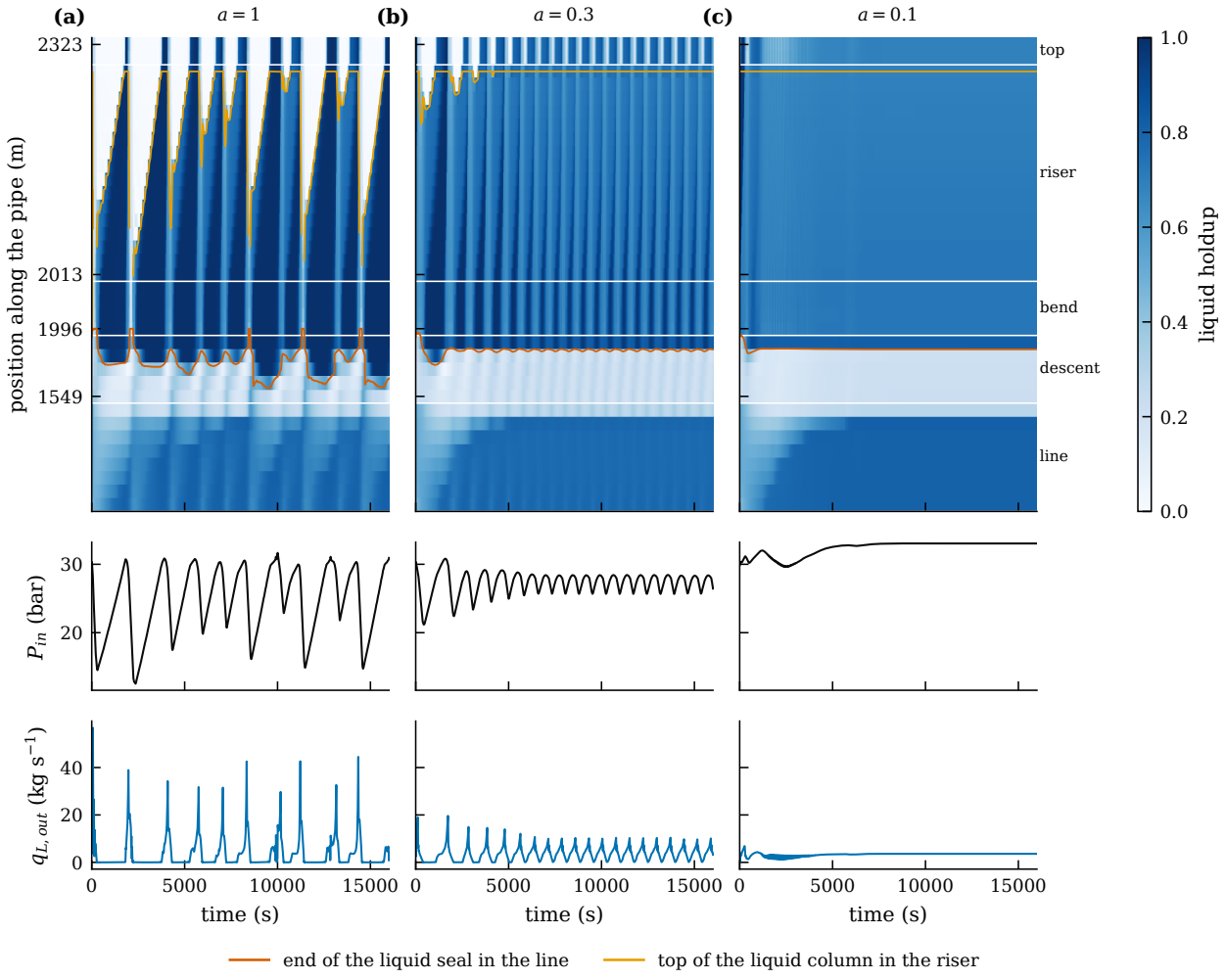


Figure 6: Liquid holdup along the flowline–riser pipe (one row per cell) against time for three choke openings at the base point, from the blocked start at $t = 0$, with the flowline inlet pressure and the liquid flow rate through the choke.

Table 8
Signature of the cycles at the base point against the choke opening (last 8000 s of each run).

opening	class	period (s)	P_{in} (bar)	$\bar{q}_L$ (kg/s)	peak/mean	seal (m)	min. riser holdup	fall of P_{in} (s)	riser refill
1	severe slugging	1530	14.74–31.66	3.861	11.5	389	0.22	428	
0.6	severe slugging	1238	21.70–30.60	3.739	8.6	215	0.54	320	
0.3	severe slugging	729	25.66–28.45	3.828	2.7	118	0.64	342	
0.2	pressure cycle	718	28.07–28.84	3.793	1.2	105	0.68	–	
0.1	stable	–	33.03–33.05	3.600	1.0	–	0.71	–	

Base point, blocked start, last 8000 s of 16 000 s. Seal: largest length of liquid-filled line upstream of the low point. Fall of P_{in}: from its maximum to its minimum in a cycle; riser refill: from the end of a blowout to a full riser (level at 98 % of its height for at least 20 s), only in the cycles in which the mean liquid holdup of the riser falls below 0.5 (medians over the cycles).

the lighter well column. The dry well is different in kind: its gas never clears the riser, whose column stays liquid-continuous (holdup above 0.70) while the flowline pressure drifts between 26.5 and 30.6 bar and the liquid leaves in pulses of 11 kg s^{-1}, three times the mean, every 1760 s — a pressure cycle with liquid pulses, as at $a = 0.3$ with the base well, with the gas bubbling through a riser that never blows out. With the gas of the reservoir fluid the flowline still runs stratified during the blocked phase: at this flowline length and these rates the system stays well inside the

Table 9
Signature of the cycles against the gas–oil ratio of the well, choke open.

GOR (m^3/m^3)	w_G	mean q_L (kg/s)	class	period (s)	P_{in} (bar)	peak q_L (kg/s)	min. riser holdup
30	0.007	3.58	severe slugging	1760	26.51–30.56	10.9	0.70
50	0.029	3.86	severe slugging	1530	14.74–31.66	44.5	0.22
80	0.059	4.02	severe slugging	1543	14.55–30.75	40.5	0.17
122.13	0.099	4.28	severe slugging	1182	14.88–28.94	43.6	0.17

Choke open at the base point, blocked start, 16 000 s; signatures over the last 8000 s.

severe-slugging region of the criterion of Bøe (1981) (Section 8.1), and the end of severe slugging at high gas rates is not reached.

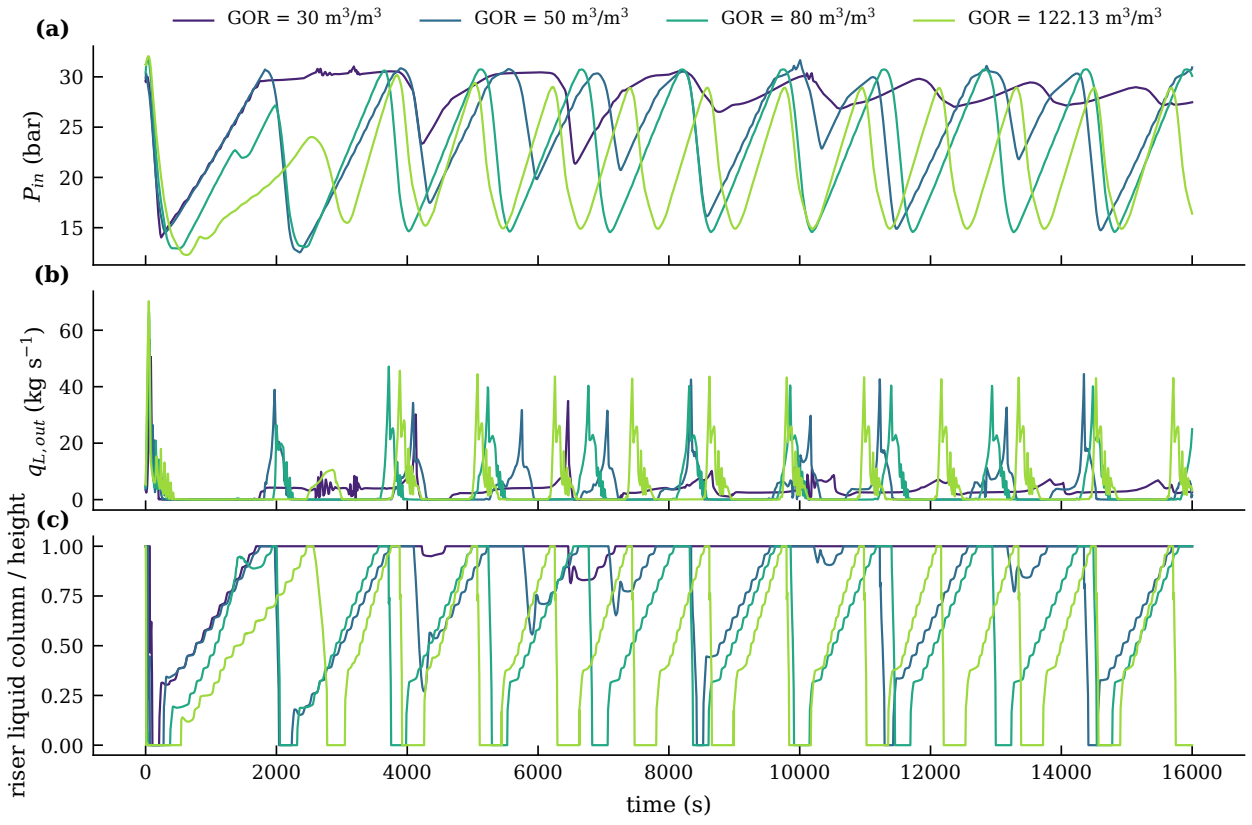


Figure 7: Four gas–oil ratios with the choke open at the base point: (a) flowline inlet pressure, (b) liquid flow rate through the choke, (c) height of the liquid column in the riser relative to the riser height.

8.5. Linear stability

Newton's method on $\mathbf{f} = \mathbf{0}$ converges on the steady state of the run at $a = 0.1$ in a few iterations, and continuation in the opening gives the steady state at every opening up to the fully open choke (Fig. 8, Table 10). Two global pairs of the generalised spectrum govern its stability. A pair with a period of about 70 s — the oscillation of the liquid column in the riser against the gas pocket of the flowline — crosses the imaginary axis at $a_H = 0.13$; a slow pair with a period of 500 to 1000 s — the exchange of liquid between the flowline and the riser, which becomes the cycle — crosses at $a \approx 0.20$, and its period grows with the opening, from 545 s at 0.2 to 965 s at 0.38, close to the 720 s of the simulated pressure cycles at 0.2 and 0.3. Above $a \approx 0.4$ the slow pair collides on the real axis and the steady state has a real unstable mode, the static runaway of the riser, whose growth rate reaches 0.046 s^{-1} with the choke open: the steady state is left within minutes and the blocked start is the natural way to the cycle. The transients agree with the spectrum:

at 0.1 the run settles on the steady state, at 0.2 it carries the 720 s pressure cycle of 0.7 bar amplitude, and from 0.3 the cycles of Table 8; the 70 s mode, although linearly unstable from 0.13, is not visible in the runs at 0.2 and 0.3. At $a_H = 0.13$ the pressure drop across the choke in the steady state is 3.7 bar, against the 20.1 bar of the liquid-filled riser; the criterion of Jansen et al. (1996), evaluated with the flowline gas fraction (0.37) and the riser holdup (0.70) of that steady state, asks for 3.5 bar — the same number — and for 5.4 bar with a riser kept full of liquid. Beyond the threshold the steady states exist and are unstable; the cycles straddle their flowline inlet pressure (Fig. 8c) and match their production on average (Fig. 8d).

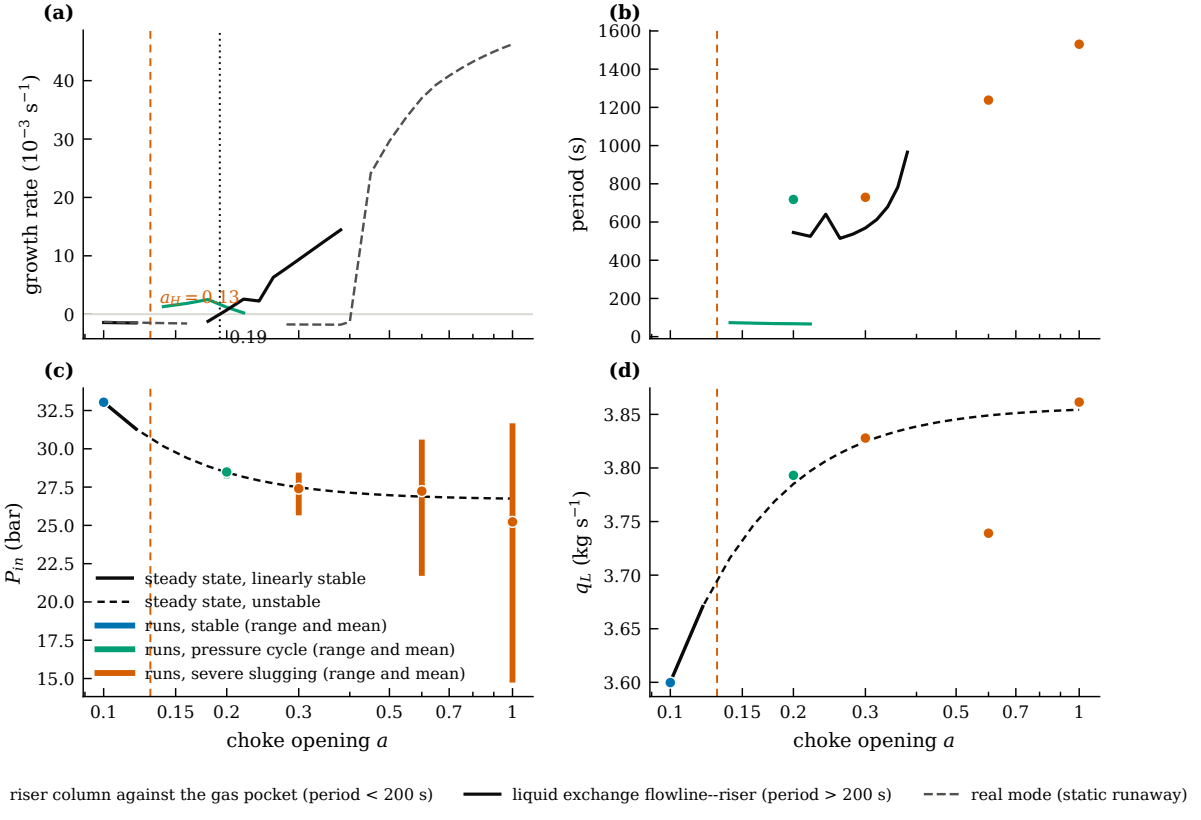


Figure 8: Steady states of the base point against the choke opening and their linear stability (generalised spectrum with the mass matrix), compared with the runs from the blocked start: (a) growth rates of the global slow modes and of the fastest of the other modes; (b) period of the slow pair and of the simulated cycles; (c) flowline inlet pressure of the steady state (solid where linearly stable, dashed where unstable) with the range and the mean of the runs; (d) liquid production of the steady state and mean of the runs. Vertical line: a_H.

8.6. Mesh dependence

Mesh dependence (runs of the campaign at 2 km pending): Table 11 compares the base mesh with a coarser one (cells twice as long) and a finer one (half as long) on the run with the choke open: whether the period, the pressure range and the liquid surge converge, and how long the seal in the flowline is in cells on each mesh.

9. Practical recommendations

The difficulties met along the way are common to transient two-fluid codes; the following recommendations summarise what was measured.

1. A numerical method that does not require hyperbolicity does not make an ill-posed model usable: ill-posedness appears as growth that increases with mesh refinement. Check the characteristics on the states the simulation visits, and regularise with physical terms where it goes.

Table 10
Steady states of the base point against the choke opening: slow pair, pressures, pressure drop across the choke and production.

a	Re λ (10^{-3} s^{-1})	period (s)	P_{in} (bar)	Δp_{choke} (bar)	q_L (kg/s)	α_G flowline	riser holdup
0.1	-1.31	–	33.04	5.80	3.60	0.36	0.71
0.14	+1.30	73	30.17	3.16	3.72	0.37	0.70
0.18	+2.52	68	28.86	1.96	3.77	0.37	0.70
0.22	+2.58	525	28.16	1.33	3.80	0.37	0.70
0.26	+6.32	514	27.75	0.96	3.81	0.37	0.70
0.3	+9.35	569	27.49	0.73	3.82	0.37	0.69
0.34	+12.07	679	27.31	0.57	3.83	0.37	0.69
0.38	+14.44	965	27.18	0.46	3.84	0.38	0.69
0.5	+29.65	–	26.97	0.27	3.85	0.38	0.69
0.7	+40.85	–	26.83	0.14	3.85	0.38	0.69
1	+46.24	–	26.75	0.07	3.85	0.38	0.69

Steady states of the base point against the choke opening (Newton with continuation in a from the end state of the run at $a = 0.1$); slow pair of the generalised spectrum, pressure at the flowline inlet, pressure drop across the choke, liquid production, length-weighted gas fraction of the flowline and liquid fraction of the riser. Two global pairs cross the imaginary axis: the oscillation of the riser column against the gas pocket (period about 67 s) at $a_H = 0.130$ and the exchange of liquid between flowline and riser (period 560 s there) at $a = 0.192$ (interpolated in $\log a$). At a_H the pressure drop across the choke is 3.65 bar against a liquid-filled riser head $\rho_L g H = 20.1$ bar, and the criterion of Jansen et al. (1996) with the flowline gas fraction and the riser holdup of that steady state ($\alpha_p = 0.6$) asks for 3.5 bar (5.4 bar with $\Phi = 1$).

Table 11
Mesh dependence of the cycle with the choke open.

mesh	cells	cell size (m)	steps	CPU (s)	interval (s)	P_{in} (bar)	peak q_L (kg/s)
base (15 + 57 cells)	57	2.0–200.0	62 025	3188	4320	13.93–33.45	40.5
coarse (8 + 32 cells)	32	2.0–384.6	59 301	2207	3312	25.73–34.60	14.7
fine (30 + 112 cells)	–	–	–	–	–	–	–

Choke open at the base point, blocked start; cells of the flowline–riser pipe; base run from the openings batch.

2. Collocated schemes without dissipation on the void waves, and backward differences applied to all variables, cannot carry the fronts of a severe-slugging cycle; a staggered scheme with phase-wise donor cells can.
3. Use the same void fraction in every term of a face momentum balance; mixing the donor value with the mean creates an odd–even mode.
4. Do not limit the void fraction to keep it in bounds; limit the fluxes of the phase that the donor cell does not have, and relax the vanishing phase towards no slip, not towards rest.
5. Integrate conservative variables. Verify the mass balance with the boundary fluxes integrated by the same recurrence as the states, never by quadrature of the outputs.
6. Flow-pattern criteria built for steady flow must be evaluated on the local state, with smooth switches of relative width and weights that form a partition of unity, before they can be used in a transient implicit solver.
7. Closure terms that are dimensionally inconsistent may compensate each other at one scale and fail at another; check them at more than one pressure and diameter.
8. A choke law with a mass-weighted mixture density does not resist gas and is singular in reverse flow; use the no-slip volume-weighted density.
9. A closed-box integrator prevents conservative residuals, projection on physical bounds and range-scaled tolerances, all of which were needed; an exact Jacobian does not reduce Newton iterations when the closures are not smooth, but compiling the right-hand side reduces the cost of each one.
10. When a transient starts on, or settles near, an equilibrium, bound the time step: an implicit multistep formula with steps much longer than the time scale of an instability damps it, and the run reports a stable state that is not. Even then, a slow instability may not grow enough within a simulated window; decide the stability of an equilibrium from its eigenvalues.
11. Linearise a conservative formulation with its mass matrix, $\mathbf{J}\mathbf{v} = \lambda\mathbf{M}\mathbf{v}$. The eigenvalues of $\mathbf{J}$ alone show fast unstable modes that do not exist and hide the slow pair whose crossing of the imaginary axis marks the onset

of the cycles; with the mass matrix the steady state predicts the onset and the period of the cycles near it, and transient simulations are needed for the amplitude of the cycles and where they coexist with a stable steady state.

10. Conclusions

A four-equation two-fluid model with the flow pattern identified dynamically along the whole system simulates severe slugging in a well–pipeline–riser system once five conditions are met. The closures must be regularised where the simulation goes: with a low-void rule, an objective virtual-mass term and an interfacial pressure in the intermittent pattern the model is hyperbolic in all but 0.9 % of the states that a severe-slugging cycle visits. The discretisation must be staggered, with phase-wise donor cells, donor face closures and a donor-cell limiter of the mass fluxes: collocated schemes do not reach the first blowout, and mean face closures create an odd–even steady state and an unstable oscillation of the cells at the bend. The time integration must conserve the phase masses: the primitive form creates 1.7 % of the liquid and loses 1.9 % of the gas in one cycle of the field case and changes the following cycles. The integrator must project the states on their bounds, scale its tolerances with the ranges of the states and bound its step near an equilibrium. And the flow-pattern criteria must be evaluated on the local state with smooth switches. A compiled right-hand side with exact Jacobians makes each step X times cheaper and the maps of hundreds of long transients affordable.

The linearisation of the conservative formulation requires its mass matrix. With it, the steady state of the discretised system predicts the onset of the cycles: a slow complex pair, which exchanges liquid between the riser and the gas pocket of the line, crosses the imaginary axis at a choke opening that grows with the reservoir pressure, and its period is that of the simulated cycles near the onset; further from it a real global mode, the static runaway of the riser, becomes unstable and the cycle becomes a relaxation cycle with complete blockage of the riser. Conclusions on the offshore case: the opening that stabilises the riser and the pressure drop across the choke at that opening, compared with the hydrostatic head of the riser and with the criterion of Jansen et al. (1996); the production penalty; the effect of the gas–oil ratio.

Acknowledgements

The authors thank the Programa de Pós-Graduação em Tecnologia de Processos Químicos e Bioquímicos of the Escola de Química, Universidade Federal do Rio de Janeiro.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools for code development and language editing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The simulator used in this study is not publicly available. Appendix C lists the geometry, every parameter of the model, the boundary conditions, the numerical scheme and the integrator, the settings of every run and the definitions of the derived quantities, so that the results can be reproduced with an independent implementation; the case definitions and the simulation results are available from the corresponding author upon reasonable request.

A. Closure relations

Wall and interfacial shear stresses are $\tau = f\,\rho v|v|/2$, with $|v|$ regularised as $(v^2+\epsilon_v^2)^{1/2}$, $\epsilon_v = 10^{-4}\,\mathrm{m\,s^{-1}}$, so that they are smooth at rest, and the Fanning factor of Churchill (1977) on a hydraulic diameter D_h,

$$f = 2\left[\left(\frac{8}{Re}\right)^{12} + \left(c_1+c_2\right)^{-3/2}\right]^{1/12}, \quad c_1 = \left[2.457\ln\frac{1}{(7/Re)^{0.9}+0.27\,\varepsilon/D_h}\right]^{16}, \quad c_2 = \left(\frac{37530}{Re}\right)^{16}, \tag{A.1}$$

with $Re = \rho D_h|v|/\mu$ and ε the wall roughness. The forces per unit length in Eq. (6) are $\Gamma_L = \tau_{wL}S_L - \tau_i S_i$ and $\Gamma_G = \tau_{wG}S_G + \tau_i S_i$, with the interfacial stress evaluated on the slip $v_r = v_G - v_L$. The acceleration of gravity in the closures is $g = 9.81\,\mathrm{m\,s^{-2}}$.

Stratified flow The wetted angle β follows from $\beta - \sin\beta = 2\pi\alpha_L$ and the level from $h_L = D[1 - \cos(\beta/2)]/2$. The perimeters are $S_L = \beta D/2$, $S_G = \pi D - S_L$ and $S_i = D\sin(\beta/2)$, the hydraulic diameters $D_L = 4A_L/S_L$ and $D_G = 4A_G/(S_G + S_i)$, and the wall factors are Eq. (A.1) at (v_k, D_k). The interfacial stress is $\tau_i = f_i\rho_G v_r|v_r|/2$ with $f_i = f_G\,[1+15\,(h_L/D)^{1/2}\max(0, |v_{GS}|/v_{GT}-1)]$ and $v_{GT} = 5.5\,\mathrm{m\,s^{-1}}$ (Andritsos and Hanratty, 1987). The pressure corrections are the hydrostatic heads between the interface and the centroids of the two phase areas, measured from the pipe bottom,

$$\Delta P_L = \alpha_G\,\Delta P_{LG}, \quad \Delta P_G = -\alpha_L\,\Delta P_{LG}, \quad \Delta P_{LG} = \left[(\rho_L - \rho_G)h_L - \rho_L z_L + \rho_G z_G\right] g\cos\theta, \tag{A.2}$$

$$z_L = D\left[\frac{1}{2} - \frac{\sin^3(\beta/2)}{3\pi\alpha_L}\right], \qquad z_G = D\left[\frac{1}{2} + \frac{\sin^3(\beta/2)}{3\pi\alpha_G}\right], \tag{A.3}$$

whose derivative with respect to α_G is the level-gradient term that keeps stratified flow hyperbolic.

Annular flow A liquid film of thickness $\delta = D(1 - \alpha_G^{1/2})/2$ wets the whole wall: $S_L = \pi D$, $S_G = 0$ and $S_i = \pi D\alpha_G^{1/2}$, with $D_L = 4A_L/S_L$ and $D_G = 4A_G/S_i$. The film factor follows Kowalski (1987), $f_L = \max[24/Re_L,\ 0.263\,\alpha_L^{-1}(D_L/(Re_L D))^{1/2}]$ with $Re_L = \rho_L D_L|v_L|/\mu_L$, and the interfacial factor is the stratified one with $h_L = \delta$. Both phases are at the mixture pressure ($\Delta P_k = 0$).

Bubbly flow The bubble diameter is $d_b = \min(d_e, d_{max})$, with $d_e = (16\sigma/g\Delta\rho)^{1/2}$ and the breakup diameter $d_{max} = [0.725 + 4.15|v_{GS}/v_M|^{1/2}](\sigma/\rho_L)^{3/5}(2f_M|v_M|^3/D)^{-2/5}$ (Hinze, 1955; Calderbank, 1958), where $v_M = v_{LS} + v_{GS}$ and f_M is Eq. (A.1) with the liquid properties at v_M on D. The terminal velocity $v_b = (2.14\sigma/\rho_L d_b + 0.505\,g d_b)^{1/2}$ (Mendelson, 1967; Fan and Tsuchiya, 1990) sets the drag coefficient $C_D = 4gd_b\Delta\rho/(3\rho_L v_b^2)$. With $S_i = \pi D^2\alpha_G/d_b$ and $\tau_i = \frac{3}{8}C_D\,\rho_L v_r|v_r|/2$ the interfacial force per unit length is $\frac{3}{4}C_D\alpha_G\rho_L A\,v_r|v_r|/d_b$, which balances buoyancy at the terminal velocity. Only the liquid has wall friction, over $S_L = \pi D\alpha_L$ with Eq. (A.1) at (v_L, D). The liquid pressure correction is $\Delta P_L = 0.25\rho_L v_r^2$ (Stuhmiller, 1977), with $|v_r|$ limited to $3\,\mathrm{m\,s^{-1}}$, and $\Delta P_G = 0$.

Intermittent flow A slug unit consists of an aerated liquid slug, a fraction r of the unit, with gas fraction α_s, and a film region, the fraction $1 - r$, with gas fraction α_f (Taitel and Barnea, 1990); the slug is treated with the bubbly closure and the film region with the stratified one, and each closure term is the average $rX_s + (1-r)X_f$. The correlations are written for co-current flow and use $u_{LS} = |v_{LS}|$, $u_{GS} = |v_{GS}|$ and $u_M = u_{LS} + u_{GS}$; the local velocities take back the sign of the superficial velocity of their phase. The slug gas fraction follows Barnea and Brauner (1985), limited to the gas content of the cell,

$$\alpha_s = \min(\alpha_s^0, \alpha_G), \qquad \alpha_s^0 = 0.058\left[2\left(\frac{0.4\,\sigma}{g\Delta\rho}\right)^{1/2}\left(\frac{2f_S u_M^3}{D}\right)^{2/5}\left(\frac{\rho_L}{\sigma}\right)^{3/5} - 0.725\right]^2, \qquad 0 \le \alpha_s^0 \le 0.52, \tag{A.4}$$

with f_S from Eq. (A.1) for the liquid at u_M on D, and the translational velocity is $v_t = 1.2\,u_M + (0.35\sin\theta + 0.54\cos\theta)(gD)^{1/2}$ (Bendiksen, 1984). The gas fraction of the film region comes from the physics of the film rather than from a slug-length correlation: in horizontal and downward sections it is the stratified equilibrium α_f^{st} of Taitel and Dukler (1976) at the superficial velocities of the cell (the same equilibrium used by the flow-pattern criteria of Appendix B); in upward sections the film falls around a Taylor bubble, and α_f^{TB} solves the kinematics of the unit with the free-falling film of Brotz (Taitel et al., 1980),

$$\left[v_t + c\,(1 - \alpha^{1/2})^{1/2}\right](1 - \alpha) = v_t(1 - \alpha_G) - u_{LS}, \qquad c = 9.916\,(gD\max(\sin\theta, 0))^{1/2}; \tag{A.5}$$

the two blend over the first 10° of upward inclination, $\alpha_f^{eq} = (1 - w)\alpha_f^{st} + w\,\alpha_f^{TB}$ with $w = u^2(3 - 2u)$, $u = \min(1, \max(0, \theta/10°))$. The film never holds more than $\alpha_{f,max} = 0.95$ of gas (beyond that the cell is annular, which

the pattern weights cover), and never less than the cell,

$$\alpha_f = \max\left[\min(\alpha_f^{eq}, \alpha_{f,max}),\ \alpha_G\right], \qquad r = \frac{\alpha_f - \alpha_G}{\alpha_f - \alpha_s} \in [0,1], \tag{A.6}$$

so that the unit conserves the gas of the cell, $\alpha_G = r\alpha_s + (1-r)\alpha_f$, reduces to a liquid slug at the state of the cell when $\alpha_G \le \alpha_s^0$ and to a film region at the state of the cell when $\alpha_G \ge \alpha_f^{eq}$. The local velocities follow from the mass balances of the unit: in the slug $u_L = [u_{LS} + v_t(\alpha_G - \alpha_s)]/(1-\alpha_s)$ and $u_b = [u_{GS} - v_t(\alpha_G - \alpha_s)]/\alpha_s$, in the film region $u_f = v_t - (v_t - u_L)(1-\alpha_s)/(1-\alpha_f)$ and $u_G = v_t - (v_t - u_b)\,\alpha_s/\alpha_f$; they return the superficial velocities of the cell, $u_{LS} = r(1-\alpha_s)u_L + (1-r)(1-\alpha_f)u_f$ and likewise for the gas. The slug front picks up the slower film and accelerates it, a loss per unit of $\Delta p_a = \rho_L(1-\alpha_s)(\alpha_f - \alpha_s)(v_t - u_L)^2/(1-\alpha_f)$ (Taitel and Barnea, 1990); with the frequency of units passing a point, ω_s/v_t per metre, it adds $A\,\Delta p_a\,\omega_s/v_t$ to the force on the liquid, against its direction, with the slug frequency of Gregory and Scott (1969) and the inclination factor of Zabaras (2000),

$$\omega_s = 0.0226\left[\frac{u_{LS}}{gD}\left(\frac{19.75}{u_M} + u_M\right)\right]^{1.2}\left[0.836 + 2.75\,\max(\sin\theta, 0)^{1/4}\right] \quad \text{(SI units)}, \tag{A.7}$$

applied with the horizontal factor in downward pipes. The loss vanishes smoothly when the unit degenerates (r within 0.05 of 0 or 1), since without a film or without a slug there is no front.

B. Flow-pattern criteria

The criteria are evaluated at every cell with the local state: the level $\tilde{h} = h_L/D$ and the angle β of the cell's void fraction, and α_G, α_L of the cell. The Froude numbers are $F_L = [\rho_L/(\Delta\rho\, g D)]^{1/2}|v_{LS}|$ and $F_G = [\rho_G/(\Delta\rho\, g D)]^{1/2}|v_{GS}|$, with $g = 9.8\,\mathrm{m\,s^{-2}}$ ($9.81\,\mathrm{m\,s^{-2}}$ elsewhere), and f_S and f_L are Eq. (A.1) for the liquid at v_{LS} and at v_{LS}/α_L on D. Each comparison $X > X_c$ is the logistic function

$$\sigma(X, X_c) = \left[1 + \exp\left(-k\,\frac{X - X_c}{|X_c|}\right)\right]^{-1}, \qquad k = 20, \tag{B.1}$$

and the direction of the mixture is $s = \frac{1}{2}[1 + \tanh(v_M/v_s)]$ with $v_s = 0.02\,\mathrm{m\,s^{-1}}$. A pipe is horizontal within 10° of the horizontal and vertical within 10° of the vertical. The criteria are

Kelvin–Helmholtz, $s = 1$:
$$\frac{F_G}{\cos^{1/2}\theta} > \left[\frac{\pi(1-\tilde{h})^2\alpha_G^3}{4\sin(\beta/2)}\right]^{1/2}, \tag{B.2}$$

film trajectory, $s = 0$:
$$\left(\frac{f_S F_L^2}{\cos\theta}\right)^{1/2} > \frac{(1-\tilde{h})^{1/2}\alpha_L}{(f_L/f_S)^{1/2}}, \tag{B.3}$$

wind waves, $s = 1$: $F_G\left(\dfrac{Re_{SL}}{\cos\theta}\right)^{1/2} > 20\,\alpha_L^{1/2}\alpha_G$, gravity waves, $s = 0$: $F_L > 1.5\,\tilde{h}^{1/2}\alpha_L$, (B.4)

film bridging:
$$\alpha_L^{film} < 0.35, \qquad \alpha_L^{film} = \max(\alpha_L^{eq}, \alpha_L), \tag{B.5}$$

annular Froude, vertical:
$$F_G > \left\{\frac{75}{Eo_D^2}\left[\left(1 + \frac{(Eo_D/75)^2}{0.0065}\right)^{1/2} - 1\right]\right\}^{1/2}, \quad Eo_D = D\left(\frac{\Delta\rho\, g}{\sigma}\right)^{1/2}, \tag{B.6}$$

bubble packing:
$$\alpha_b < 0.25\ \text{(bubbly)}, \quad \alpha_b > 0.52\ \text{(intermittent)}, \tag{B.7}$$

dispersed bubbles:
$$d_{max} < d_{def} = 2\left(\frac{0.4\,\sigma}{\Delta\rho\, g}\right)^{1/2}, \quad d_{max} < d_{migr} = \frac{3}{8}\frac{\rho_L}{\Delta\rho}\frac{f_M v_M^2}{g\cos\theta}, \tag{B.8}$$

turbulent dispersion, horizontal:
$$\left(\frac{2 f_S F_L^2}{\cos\theta}\right)^{1/2} > \left[\frac{2\pi\alpha_L^2\alpha_G}{\sin(\beta/2)\,(f_L/f_S)}\right]^{1/2}, \tag{B.9}$$

Table C.1
Geometry and mesh of the application.

pipe: section	D (mm)	start s (m)	length (m)	end x (m)	end z (m)	inclination (deg)	cells
well: well	127	0.0	1500.0	0.0	-300.0	+90.0	15
flowline–riser: flowline, horizontal	152.4	0.0	5000.0	5000.0	-300.0	+0.0	25
flowline–riser: flowline, descent	152.4	5000.0	994.8	5994.2	-334.7	-2.0	10
flowline–riser: bend, chord 1	152.4	5994.8	2.0	5996.2	-334.4	+9.5	1
flowline–riser: bend, chord 2	152.4	5996.8	2.0	5997.9	-333.3	+32.5	1
flowline–riser: bend, chord 3	152.4	5998.8	2.0	5999.0	-331.7	+55.5	1
flowline–riser: bend, chord 4	152.4	6000.8	2.0	5999.4	-329.7	+78.5	1
flowline–riser: riser	152.4	6002.8	329.7	5999.4	0.0	+90.0	16
flowline–riser: top section	152.4	6332.5	10.0	6009.4	0.0	+0.0	2

Coordinates with x horizontal and z the elevation above sea level; the well goes from its bottom to the wellhead at the start of the flowline. The bend is a circular arc of 5 m radius represented by four chords. Wall roughness: 0.2 mm in the well, 0.6 mm in the flowline and riser. Mesh variants (Section 8.6): coarse, 8 cells in the well and 13, 5, 4, 8, 2 in the sections of the flowline–riser pipe; fine, 30 in the well and 50, 20, 8, 32, 2.

where $Re_{SL} = \rho_L D|v_{LS}|/\mu_L$, α_L^{eq} is the film fraction of the annular equilibrium of the superficial velocities (Taitel et al., 1980; Barnea, 1987), α_b is the smooth maximum of the drift-flux estimate $|v_{GS}|/(|1.25\,v_M + u_b| + 10^{-3})$, $u_b = 1.53(\sigma g \Delta\rho/\rho_L^2)^{1/4}\sin\theta$, and the cell's α_G, $\max_\varepsilon(a,b) = [a+b+((a-b)^2+\varepsilon^2)^{1/2}]/2$ with $\varepsilon = 0.02$, and d_{max} is the breakup diameter of the bubbly closure with f_M evaluated at the density of the drift-flux mixture. With σ_{KH}, σ_{tr}, … the logistic values of the criteria, the weights of Eq. (9) are

$$w_S = \left\{(1-\sigma_{KH})\,s + \left[1-\sigma_{tr}(1-s)\right](1-s)\right\}(1-\chi_V), \qquad p_A = 1-(1-p_A^0)(1-\sigma(\alpha_G, 0.9)), \tag{B.10}$$

$$p_A^0 = (1-\sigma_{br})\left[(1-\chi_V) + \sigma_{Fr}\,\chi_V\right], \qquad p_B = \left[\sigma_{<0.25} + \sigma_{disp}(1-\sigma_{>0.52})^2(1-\sigma_{<0.25})^2\right](1-\chi_H) + \sigma_T\,\chi_H, \tag{B.11}$$

with χ_H and χ_V the horizontal and vertical indicators, $\sigma_{br} = \sigma(\alpha_L^{film}, 0.35)$ and σ_{disp} the product of the two dispersed-bubble criteria; the wavy part of stratified flow has the weight $w_S[\sigma_K s + \sigma_{F_L}(1-s)]$. Below $\alpha_G = 0.02$ the weights blend into the bubbly ones with $w = u^2(3-2u)$, $u = \min[1, \max(0, (2\alpha_* - \alpha_G)/\alpha_*)]$ and $\alpha_* = 0.01$ (low-void rule).

C. Data for reproduction

The tables of this appendix were generated from the input files and the code of the simulator, so that they describe exactly what was run, and they are meant to allow the results to be reproduced with an independent implementation. Table C.1 gives the geometry and the mesh, Table C.2 every parameter of the model, the boundary conditions, the scheme and the integrator, Table C.3 the runs behind each section, Table C.4 the changes of each variant of the ablation, and Table C.5 the definitions of the quantities derived from the runs. The results were obtained with Python 3.14.7, NumPy 2.5.4, SciPy 1.18.1 and JAX 0.11.2.

Table C.2
Parameters of the application case.

group	item	value
fluids	liquid	$\rho_L = 646.7\,(P/P_0)^{1/\gamma_L}$ kg/m^3, $\gamma_L = 1000$, $\mu_L = 0.000\,314$ Pa s
	gas	$\rho_G = 15\,(P/P_0)^{1/\gamma_G}$ kg/m^3, $\gamma_G = 1$, $\mu_G = 1.21 \times 10^{-5}$ Pa s; $P_0 = 20$ bar
	surface tension	$\sigma = 0.0163$ N/m
	PVT (standard conditions pvt['std'])	$R_s = 23.24$ and $\mathrm{GOR}_{fluid} = 122.13$ m^3/m^3 at 20 bar and 40 °C; $\rho_o = 673.25$, $\rho_{gd} = 1.1547$ $\rho_{gf} = 0.7763$ kg/m^3
well	inflow	$J = 5$ m^3/(d bar), GOR $= 50$ m^3/m^3, $P_{res} = 200$ bar; $w_G = 0.02882$, $J_L = 0.04051$ $J_G = 0.001\,202$ kg/(s bar); $q_k = J_k \max(0, P_{res} - P_{wf})$
	choke	$\Delta P = \dot{m}Q/(Ka)^2$ [bar], $K = 0.0075\,c_v$, $c_v = 150$ ($Ka = 35.58$ cm^2 at $a = 1$); $\lvert Q\rvert$ regularised with 10^{-6} m^3/s
	separator	10 bar
	wellhead	connector: the pressure of the flowline inlet is imposed on the well outlet, the well flow rates on the flowline inlet
	pipe ends	well inlet: flow rates from the inflow law, zero gradient of α_G; riser outlet: pressure from the choke law
	intermittent	slug unit with the film at equilibrium, Zabaras frequency and Stuhmiller interfacia pressure with the cell slip (Appendix A)
closures	evaluation bounds	α_G cut to [0.001, 0.999] inside the closures
	stratified	interfacial factor of Andritsos and Hanratty with $v_{GT} = 5.5$ m/s
	bubbly	physical drag (terminal velocity with g and σ/ρ_L); $\Delta P_L = 0.25\rho_L v_r^2$, $\lvert v_r\rvert \le 3$ m/s
	virtual mass	$C_{vm} = 0.5$, $\lambda = 1$, weighted by the bubbly weight w_B
	interfacial pressure	Bestion term off ($\delta = 0$)
flow patterns	identification	automatic, smooth and local; logistic switches with $k = 20$ (Appendix B)
	low-void rule	bubbly closures below $\alpha_G = 0.01$, C^1 ramp up to 0.02
	local rules	annular above $\alpha_G = 0.9$; no stratified within 10° of vertical; film bridging with the cel liquid fraction
scheme	staggered FV	conservative variables ($\rho_G\alpha_G, \rho_L\alpha_L$); face closures with the donor fraction
	donor switch	$w = \frac{1}{2}[1 + \tanh(q/\epsilon)]$, $\epsilon = 0.5 \max \lvert q\rvert_{\mathrm{nb}} + q_s$, $q_{s,L} = 0.0405$, $q_{s,G} = 0.0012$ kg/s
	flux limiter	C^1 ramp of the donor phase fraction over [0, 0.001], donor switch with $\epsilon = 0.05\,q_s$
	vanishing phase	relaxed to the no-slip flow rate with $\tau = 0.05$ s
	artificial viscosity	$\nu = Cch$, $C = 0.05$
	initial state	blocked start of the case generator (Section 8.1), stored at the nodes of the scheme
integrator	formula	BDF of order 1–2, variable step, modified Newton
	tolerances	rtol = 0.003; atol = 0.01 × (1, 35.36 bar, 20.3 kg/s, 0.601 kg/s) for (α_G, P, q_L, q_G)
	Newton	tolerance 0.001 (scaled), at most 20 iterations
	steps	first 0.001 s, minimum 1×10^{-9} s, maximum none (1 s from an equilibrium)
	projection	$\alpha_G \in [-0.0005, 1.005]$, order 1 after a projection
	backend	JAX, exact Jacobian; boundary fluxes integrated with the same recurrence

Values read from the case file of the application and from the simulator (Section 8.1).

Table C.3
Runs behind each section.

study (section)	start	time (s)	output (s)	max. step (s)	budget (s)	runs
choke openings (8.3)	blocked	16000	2.0	none	none	5
gas–oil ratio (8.4)	blocked	16000	2.0	none	none	4
ablation (6.3)	blocked	8000	2.0	none	none	16
mesh (8.6)	blocked	8000	2.0	none	none	2
backends (7)	blocked	2000	2.0	none	none	2
Newton iteration (6.1)	states of the open-choke run	150	2	none	none	18
model problem (2)	steady state + 0.1 %	300	0.1	0.5	none	4

Each run is the application case of Table C.2 with the changes of its study (choke opening, gas–oil ratio, mesh, or the variants of Table C.4). Budget: wall-clock time after which a run is stopped with its partial trajectory. The model problem uses rtol = 10^{-4} and atol = 10^{-6} (dimensionless units).

Table C.4
Variants of the ablation (Table 4) as changes to the case file of the application.

variant	change to the configuration of the application
reference (Section 8)	–
primitive time integration	the integrator advances the primitive variables (α_G, P) instead of the masses unit volume; vanishing phase relaxed to a zero flow rate (the no-slip target ne the flux limiter of the conservative form)
no acoustic viscosity	$C = 0$ in the artificial viscosity
face closures with the mean void fraction	face closures evaluated with the arithmetic mean of the void fractions of the neighbouring cells
no low-void rule	low-void threshold 10^{-12} (rule inactive)
no virtual mass	$C_{vm} = 0$
bubble drag without g and σ/ρ_L	bubble terminal velocity $v_b = (0.51\, d_b + 2.14/d_b)^{1/2}$, without g and σ/ρ_L
slug unit of the thesis, mass-consistent	slug unit of the thesis with the slug shortened so that the unit conserves mass of the cell (film cut at $\alpha_f = 0.95$)
slug unit of the thesis	slug unit with the slug length of Norris and the film holdup from the m balance, cut at $\alpha_f = 0.95$
slug frequency of the thesis	slug frequency $\omega_s = (u_{GS}/D)[0.0323/2\,(u_{LS}/u_{GS})(\rho_L/\Delta\rho)^{1/2} f_L^{-1/2} Eo^{-0.2}]^n$, $N_f/(260 + 0.85 N_f)$
logistic switches with $k = 80$	logistic switches with $k = 80$ instead of 20
bubble criteria on the mixture flow rates (no local void fraction)	the bubbly-flow criteria of the pattern identification evaluated on the v fraction implied by the phase flow rates instead of the local void fraction the cell
first-order BDF	BDF of order 1 only
single absolute tolerance 10^{-6}	absolute tolerance 10^{-6} for every state, with the pressure in MPa and the f rates in kg/s
Newton without line search	Newton iteration without the line search: the step is halved only when the resid grows, never in the first iteration
first-order error estimate (linear predictor)	local error estimated from the linear predictor, $r/(1 + 2r)\,(\mathbf{x} - \mathbf{x}^{pred})$, instead the quadratic predictor with the Milne constant

Every variant has the choke wide open ($a = 1$) at the base point and runs for $8000\,\mathrm{s}$.

Table C.5
Definitions of the quantities derived from the runs.

quantity	definition
cycle class	stable if the range of the smoothed P_{in} is below max(0.1 bar, 1 % of its mean) or it crosses its mean at most once; severe slugging if the liquid outflow peaks at 2 times its mean or more; pressure cycle otherwise
smoothing, period	centred moving average of 20 s on P_{in}; period = mean time between up-crossings of the mean
windows	dense runs: last $3000\,\mathrm{s}$; map: windows of $2000\,\mathrm{s}$, stop when the range is below $0.1\,\mathrm{bar}$ or two windows agree within $10\,\%$ in range and period, at most $8000\,\mathrm{s}$
blowdowns (tables)	minima of P_{in} below the middle of its 2–98 % percentile range, after $300\,\mathrm{s}$, at least $150\,\mathrm{s}$ apart
fronts	seal: liquid-filled length upstream of the low point ($\alpha_G < 0.5$); level: top of the liquid column in the riser ($\alpha_G < 0.5$); riser full: level above 98 % of the riser for $20\,\mathrm{s}$; blowout start: maximum of the smoothed P_{in}
linear stability	eigenvalues of $\mathbf{J}\mathbf{v} = \lambda\mathbf{M}\mathbf{v}$ with the exact Jacobian; global slow modes: $\lvert\lambda\rvert < 0.2\ \mathrm{s}^{-1}$ and $n_{\mathrm{eff}} \geq 4$, $n_{\mathrm{eff}} = 1/\sum_j p_j^2$ with p_j the share of node j in $\lVert\mathbf{v}\rVert^2$, states scaled by their tolerance ranges; class from the leading global slow mode
continuation	Newton on $\mathbf{f} = 0$ (residual 10^{-9}, at most 60 iterations) from the neighbouring opening, seeded by the end state of the run at $a = 0.1$ brought to the exact steady state; openings 0.10–0.38 in steps of 0.02 and 0.40–1.0 in steps of 0.05, with geometric bisection of the step (up to five levels) when Newton fails; a_H by linear interpolation of Re λ in $\log a$
choke criterion	Δp_c of Jansen et al. (1996) (their eq. 10) with α the length-weighted gas fraction of the flowline and Φ the liquid fraction of the riser of the steady state at a_H, $\alpha_p = 0.6$, L the flowline length, H the riser height and P_S the separator pressure; choke Δp = pressure at the top of the riser minus P_S in the steady state
well-posedness	eigenvalues of $\mathbf{A}$ in every cell, one state every $5\,\mathrm{s}$; ill-posed if the largest $\lvert\mathrm{Im}\,\lambda\rvert > 10^{-6}$ m/s
mass error	inventory change minus the boundary fluxes integrated with the recurrence of the integrator

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